

Emotions and Motivation

Affective Neuroscience

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This module provides a brief overview of the neuroscience of emotion. It integrates findings from human and animal research to describe the brain networks and associated neurotransmitters involved in basic affective systems.

Learning Objectives

- Define affective neuroscience.
- Describe neuroscience techniques used to study emotions in humans and animals.
- Name five emotional systems and their associated neural structures and neurotransmitters.
- Give examples of exogenous chemicals (e.g., drugs) that influence affective systems, and discuss their effects.
- Discuss multiple affective functions of the amygdala and the nucleus accumbens.
- Name several specific human emotions, and discuss their relationship to the affective systems of nonhuman animals.

Affective Neuroscience: What is it?

Affective neuroscience examines how the brain creates emotional responses. Emotions are psychological phenomena that involve changes to the body (e.g., facial expression), changes in autonomic nervous system activity, feeling states (subjective responses), and urges to act

in specific ways (motivations; Izard, 2010). Affective neuroscience aims to understand how matter (brain structures and chemicals) creates one of the most fascinating aspects of mind, the emotions. Affective neuroscience uses unbiased, observable measures that provide credible evidence to other sciences and laypersons on the importance of emotions. It also leads to biologically based treatments for affective disorders (e.g., depression).



Although we all experience emotions all the time, they are very difficult to describe and study. Fortunately, technological advances are making this easier. [Image: Waag Society, <https://goo.gl/F0KdnB>, CC BY-NC-SA 2.0, <https://goo.gl/iF4hmM>]

The human brain and its responses, including emotions, are complex and flexible. In comparison, nonhuman animals possess simpler nervous systems and more basic emotional responses. Invasive neuroscience techniques, such as electrode implantation, lesioning, and hormone administration, can be more easily used in animals than in humans. Human neuroscience must rely primarily on noninvasive techniques such as electroencephalography (EEG) and functional magnetic resonance imaging (fMRI), and on studies of individuals with brain lesions caused by accident or disease. Thus, animal research provides useful models for understanding affective processes in humans. Affective circuits found in other species, particularly social mammals such as rats, dogs, and monkeys, function similarly to human affective networks,

although nonhuman animals' brains are more basic.

In humans, emotions and their associated neural systems have additional layers of complexity and flexibility. Compared to animals, humans experience a vast variety of nuanced and sometimes conflicting emotions. Humans also respond to these emotions in complex ways, such that conscious goals, values, and other cognitions influence behavior in addition to emotional responses. However, in this module we focus on the similarities between organisms, rather than the differences. We often use the term "organism" to refer to the individual who is experiencing an emotion or showing evidence of particular neural activations. An organism could be a rat, a monkey, or a human.

Across species, emotional responses are organized around the organism's survival and reproductive needs. Emotions influence perception, cognition, and behavior to help

organisms survive and thrive (Farb, Chapman, & Anderson, 2013). Networks of structures in the brain respond to different needs, with some overlap between different emotions. Specific emotions are not located in a single structure of the brain. Instead, emotional responses involve networks of activation, with many parts of the brain activated during any emotional process. In fact, the brain circuits involved in emotional reactions include nearly the entire brain (Berridge & Kringelbach, 2013). Brain circuits located deep within the brain below the cerebral cortex are primarily responsible for generating basic emotions (Berridge & Kringelbach, 2013; Panksepp & Biven, 2012). In the past, research attention was focused on specific brain structures that will be reviewed here, but future research may find that additional areas of the brain are also important in these processes.

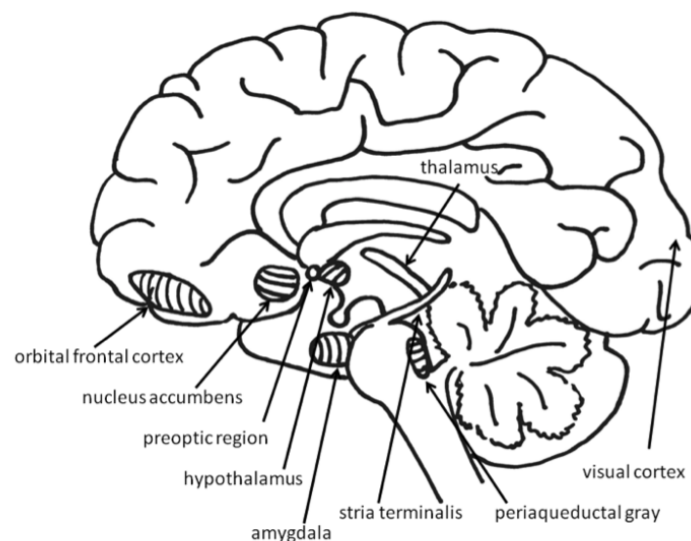


Figure 1: Structures in the brain

Basic Emotions

Desire: The neural systems of reward seeking

One of the most important affective neuronal systems relates to feelings of desire, or the appetite for rewards. Researchers refer to these appetitive processes using terms such as “wanting” (Berridge & Kringelbach, 2008), “seeking” (Panksepp & Biven, 2012), or “behavioural activation sensitivity” (Gray, 1987). When the appetitive system is aroused, the organism shows enthusiasm, interest, and curiosity. These neural circuits motivate the animal to move through its environment in search of rewards such as appetizing foods, attractive sex partners,

and other pleasurable stimuli. When the appetitive system is underaroused, the organism appears depressed and helpless.

Much evidence for the structures involved in this system comes from animal research using direct brain stimulation. When an electrode is implanted in the lateral **hypothalamus** or in cortical or mesencephalic regions to which the hypothalamus is connected, animals will press a lever to deliver electrical stimulation, suggesting that they find the stimulation pleasurable. The regions in the desire system also include the amygdala, nucleus accumbens, and **frontal cortex** (Panksepp & Biven, 2012). The neurotransmitter dopamine, produced in the mesolimbic and mesocortical dopamine circuits, activates these regions. It creates a sense of excitement, meaningfulness, and anticipation. These structures are also sensitive to drugs such as cocaine and amphetamines, chemicals that have similar effects to dopamine (Panksepp & Biven, 2012).

Research in both humans and nonhuman animals shows that the left frontal cortex (compared to the right frontal cortex) is more active during appetitive emotions such as desire and interest. Researchers first noted that persons who had suffered damage to the left frontal cortex developed depression, whereas those with damage to the right frontal cortex developed mania (Goldstein, 1939). The relationship between left frontal activation and approach-related emotions has been confirmed in healthy individuals using EEG and fMRI (Berkman & Lieberman, 2010). For example, increased left frontal activation occurs in 2- to 3-day-old infants when sucrose is placed on their tongues (Fox & Davidson, 1986), and in hungry adults as they view pictures of desirable desserts (Gable & Harmon-Jones, 2008). In addition, greater left frontal activity in appetitive situations has been found to relate to dopamine (Wacker, Mueller, Pizzagalli, Hennig, & Stemmler, 2013).



Just looking at an image of appealing food should increase the activity in your left frontal cortex. Yum! [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

“Liking” : The neural circuits of pleasure and enjoyment

Surprisingly, the amount of desire an individual feels toward a reward need not correspond to how much he or she likes that reward. This is because the neural structures involved in the enjoyment of rewards are different from the structures involved in the desire for the rewards. “Liking” (e.g., enjoyment of a sweet liquid) can be measured in babies and nonhuman animals by measuring licking speed, tongue protrusions, and happy facial expressions, whereas “wanting” (desire) is shown by the willingness to work hard to obtain a reward (Berridge & Kringelbach, 2008). Liking has been distinguished from wanting in research on topics such as drug abuse. For example, drug addicts often desire drugs even when they know that the ones available will not provide pleasure (Stewart, de Wit, & Eikelboom, 1984).

Research on liking has focused on a small area within the nucleus accumbens and on the posterior half of the ventral pallidum. These brain regions are sensitive to opioids and endocannabinoids. Stimulation of other regions of the reward system increases wanting, but does not increase liking, and in some cases even decreases liking. The research on the distinction between desire and enjoyment contributes to the understanding of human addiction, particularly why individuals often continue to frantically pursue rewards such as cocaine, opiates, gambling, or sex, even when they no longer experience pleasure from obtaining these rewards due to habituation.

The experience of pleasure also involves the orbitofrontal cortex. Neurons in this region fire when monkeys taste, or merely see pictures of, desirable foods. In humans, this region is activated by pleasant stimuli including money, pleasant smells, and attractive faces (Gottfried, O’Doherty & Dolan, 2002; O’Doherty, Deichmann, Critchley, & Dolan, 2002; O’Doherty, Kringelbach, Rolls, Hornak, & Andrews, 2001; O’Doherty, Winston, Critchley, Perrett, Burt, & Dolan, 2003).

Fear: The neural system of freezing and fleeing

Fear is an unpleasant emotion that motivates avoidance of potentially harmful situations. Slight stimulation of the fear-



Because fear is so important for our survival (i.e., fear informs us when something threatens us), our brains are able to “recognize” frightening stimuli before we ourselves are even consciously aware of them. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

related areas in the brain causes animals to freeze, whereas intense stimulation causes them to flee. The fear circuit extends from the central amygdala to the **periaqueductal gray** in the midbrain. These structures are sensitive to glutamate, corticotrophin releasing factor, adreno-cortico-trophic hormone, cholecystokinin, and several different neuropeptides. Benzodiazepines and other tranquilizers inhibit activation in these areas (Panksepp & Biven, 2012).

The role of the **amygdala** in fear responses has been extensively studied. Perhaps because fear is so important to survival, two pathways send signals to the amygdala from the sensory organs. When an individual sees a snake, for example, the sensory information travels from the eye to the **thalamus** and then to the visual cortex. The **visual cortex** sends the information on to the amygdala, provoking a fear response. However, the thalamus also quickly sends the information straight to the amygdala, so that the organism can react before consciously perceiving the snake (LeDoux, Farb, & Ruggiero, 1990). The pathway from the thalamus to the amygdala is fast but less accurate than the slower pathway from the visual cortex. Damage to the amygdala or areas of the ventral hippocampus interferes with fear conditioning in both humans and nonhuman animals (LeDoux, 1996).

Rage: The circuits of anger and attack

Anger or rage is an arousing, unpleasant emotion that motivates organisms to approach and attack (Harmon-Jones, Harmon-Jones, & Price, 2013). Anger can be evoked through goal frustration, physical pain, or physical restraint. In territorial animals, anger is provoked by a stranger entering the organism's home territory (Blanchard & Blanchard, 2003). The neural networks for anger and fear are near one another, but separate (Panksepp & Biven, 2012). They extend from the medial amygdala, through specific parts of the hypothalamus, and into the periaqueductal gray of the midbrain. The anger circuits are linked to the appetitive circuits, such that lack of an anticipated reward can provoke rage. In addition, when humans are angered, they show increased left frontal cortical activation, supporting the idea that anger is an approach-related emotion (Harmon-Jones et al., 2013). The neurotransmitters involved in rage are not yet well understood, but Substance P may play an important role (Panksepp & Biven, 2012). Other neurochemicals that may be involved in anger include testosterone (Peterson & Harmon-Jones, 2012) and arginine-vasopressin (Heinrichs, von Dawans, & Domes, 2009). Several chemicals inhibit the rage system, including opioids and high doses of antipsychotics, such as chlorpromazine (Panksepp & Biven, 2012).

Love: The neural systems of care and attachment



Just as scientists today distinguish between types of love like “romantic” and “parental,” so did the ancient Greeks who used the terms “eros” and “storge.” [Image: Go-tea 郭天, <https://goo.gl/KLFxJP>, CC BY 2.0, <https://goo.gl/zHmGV2>]

(endorphins and enkephalins).

For social animals such as humans, attachment to other members of the same species produces the positive emotions of attachment: love, warm feelings, and affection. The emotions that motivate nurturing behavior (e.g., maternal care) are distinguishable from those that motivate staying close to an attachment figure in order to receive care and protection (e.g., infant attachment). Important regions for maternal nurturing include the dorsal **preoptic area** (Numan & Insel, 2003) and the bed nucleus of the **stria terminalis** (Panksepp, 1998). These regions overlap with the areas involved in sexual desire, and are sensitive to some of the same neurotransmitters, including oxytocin, arginine-vasopressin, and endogenous opioids

Grief: The neural networks of loneliness and panic

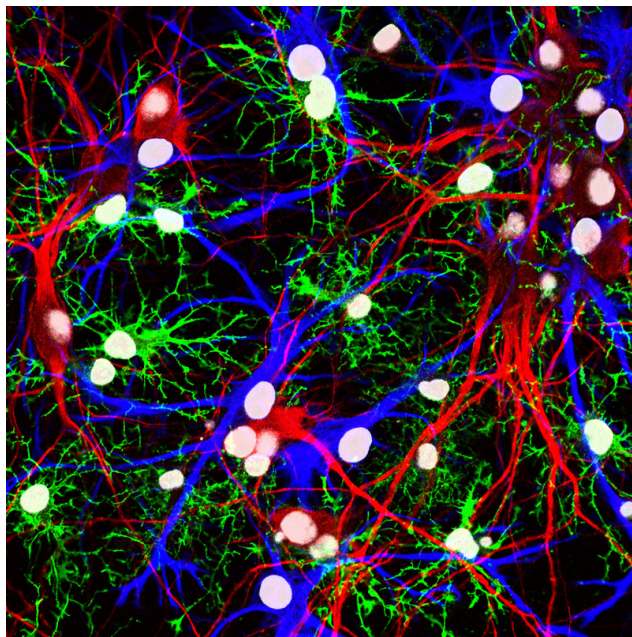
The neural networks involved in infant attachment are also sensitive to separation. These regions produce the painful emotions of grief, panic, and loneliness. When infant humans or other infant mammals are separated from their mothers, they produce distress vocalizations, or crying. The attachment circuits are those that cause organisms to produce distress vocalizations when electrically stimulated.

The attachment system begins in the midbrain periaqueductal gray, very close to the area that produces physical pain responses, suggesting that it may have originated from the pain circuits (Panksepp, 1998). Separation distress can also be evoked by stimulating the dorsomedial thalamus, ventral septum, dorsal preoptic region, and areas in the bed nucleus of stria terminalis (near sexual and maternal circuits; Panksepp, Normansell, Herman, Bishop, & Crepeau, 1988).

These regions are sensitive to endogenous opiates, oxytocin, and prolactin. All of these neurotransmitters *prevent* separation distress. Opiate drugs such as morphine and heroin, as well as nicotine, artificially produce feelings of pleasure and gratification, similar to those

normally produced during positive social interactions. This may explain why these drugs are addictive. Panic attacks appear to be an intense form of separation distress triggered by the attachment system, and panic can be effectively relieved by opiates. Testosterone also reduces separation distress, perhaps by reducing attachment needs. Consistent with this, panic attacks are more common in women than in men.

Plasticity: Experiences can alter the brain



Neural plasticity can be summed up in the phrase: "Neurons that fire together, wire together." Or in other words, when certain emotions are paired with certain contexts, we learn to associate the two together. [Image: NICHD NIH, <https://goo.gl/T3PUlg>, CC BY 2.0, <https://goo.gl/BRvSA7>]

(Reynolds & Berridge, 2008).

The responses of specific neural regions may be modified by experience. For example, the front shell of the nucleus accumbens is generally involved in appetitive behaviors, such as eating, and the back shell is generally involved in fearful defensive behaviors (Reynolds & Berridge, 2001, 2002). Research using human neuroimaging has also revealed this front-back distinction in the functions of the nucleus accumbens (Seymour, Daw, Dayan, Singer, & Dolan, 2007). However, when rats are exposed to stressful environments, their fear-generating regions expand toward the front, filling almost 90% of the nucleus accumbens shell. On the other hand, when rats are exposed to preferred home environments, their fear-generating regions shrink and the appetitive regions expand toward the back, filling approximately 90% of the shell

Brain structures have multiple functions

Although much affective neuroscience research has emphasized whole structures, such as the amygdala and nucleus accumbens, it is important to note that many of these structures are more accurately referred to as complexes. They include distinct groups of nuclei that perform different tasks. At present, human neuroimaging techniques such as fMRI are unable to examine the activity of individual nuclei in the way that invasive animal neuroscience can.

For instance, the amygdala of the nonhuman primate can be divided into 13 nuclei and cortical areas (Freese & Amaral, 2009). These regions of the amygdala perform different functions. The central nucleus sends outputs involving brainstem areas that result in innate emotional expressions and associated physiological responses. The basal nucleus is connected with striatal areas that are involved with actions such as running toward safety. Furthermore, it is not possible to make one-to-one maps of emotions onto brain regions. For example, extensive research has examined the involvement of the amygdala in fear, but research has also shown that the amygdala is active during uncertainty (Whalen, 1998) as well as positive emotions (Anderson et al., 2003; Schulkin, 1990).

Conclusion

Research in affective neuroscience has contributed to knowledge regarding emotional, motivational, and behavioral processes. The study of the basic emotional systems of nonhuman animals provides information about the organization and development of more complex human emotions. Although much still remains to be discovered, current findings in affective neuroscience have already influenced our understanding of drug use and abuse, psychological disorders such as panic disorder, and complex human emotions such as desire and enjoyment, grief and love.

Outside Resources

Video: A 1-hour interview with Jaak Panksepp, the father of affective neuroscience

<http://www.youtube.com/watch?v=u4ICY6-7hJo>

Video: A 15-minute interview with Kent Berridge on pleasure in the brain

<http://www.youtube.com/watch?v=51rGE1DgIo0>

Video: A 5-minute interview with Joseph LeDoux on the amygdala and fear

<http://www.youtube.com/watch?v=fDD5wvFMH6U>

Web: Brain anatomy interactive 3D model

<http://www.pbs.org/wnet/brain/3d/index.html>

Discussion Questions

1. The neural circuits of “liking” are different from the circuits of “wanting.” How might this relate to the problems people encounter when they diet, fight addictions, or try to change other habits?
2. The structures and neurotransmitters that produce pleasure during social contact also produce panic and grief when organisms are deprived of social contact. How does this contribute to an understanding of love?
3. Research shows that stressful environments increase the area of the nucleus accumbens that is sensitive to fear, whereas preferred environments increase the area that is sensitive to rewards. How might these changes be adaptive?

Vocabulary

Affect

An emotional process; includes moods, subjective feelings, and discrete emotions.

Amygdala

Two almond-shaped structures located in the medial temporal lobes of the brain.

Hypothalamus

A brain structure located below the thalamus and above the brain stem.

Neuroscience

The study of the nervous system.

Nucleus accumbens

A region of the basal forebrain located in front of the preoptic region.

Orbital frontal cortex

A region of the frontal lobes of the brain above the eye sockets.

Periaqueductal gray

The gray matter in the midbrain near the cerebral aqueduct.

Preoptic region

A part of the anterior hypothalamus.

Stria terminalis

A band of fibers that runs along the top surface of the thalamus.

Thalamus

A structure in the midline of the brain located between the midbrain and the cerebral cortex.

Visual cortex

The part of the brain that processes visual information, located in the back of the brain.

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Functions of Emotions

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Emotions play a crucial role in our lives because they have important functions. This module describes those functions, dividing the discussion into three areas: the intrapersonal, the interpersonal, and the social and cultural functions of emotions. The section on the intrapersonal functions of emotion describes the roles that emotions play within each of us individually; the section on the interpersonal functions of emotion describes the meanings of emotions to our relationships with others; and the section on the social and cultural functions of emotion describes the roles and meanings that emotions have to the maintenance and effective functioning of our societies and cultures at large. All in all we will see that emotions are a crucially important aspect of our psychological composition, having meaning and function to each of us individually, to our relationships with others in groups, and to our societies as a whole.

Learning Objectives

- Gain an appreciation of the importance of emotion in human life.
- Understand the functions and meanings of emotion in three areas of life: the intrapersonal, interpersonal, and social-cultural.
- Give examples of the role and function of emotion in each of the three areas described.

Introduction

It is impossible to imagine life without emotion. We treasure our feelings—the joy at a ball

game, the pleasure of the touch of a loved one, or the fun with friends on a night out. Even negative emotions are important, such as the sadness when a loved one dies, the anger when violated, the fear that overcomes us in a scary or unknown situation, or the guilt or shame toward others when our sins are made public. Emotions color life experiences and give those experiences meaning and flavor.

In fact, emotions play many important roles in people's lives and have been the topic of scientific inquiry in psychology for well over a century (Cannon, 1927; Darwin, 1872; James, 1890). This module explores why we have emotions and why they are important. Doing so requires us to understand the function of emotions, and this module does so below by dividing the discussion into three sections. The first concerns the **intrapersonal** functions of emotion, which refer to the role that emotions play within each of us individually. The second concerns the **interpersonal** functions of emotion, which refer to the role emotions play between individuals within a group. The third concerns the **social and cultural** functions of emotion, which refer to the role that emotions play in the maintenance of social order within a society. All in all, we will see that emotions inform us of who we are, what our relationships with others are like, and how to behave in social interactions. Emotions give meaning to events; without emotions, those events would be mere facts. Emotions help coordinate interpersonal relationships. And emotions play an important role in the cultural functioning of keeping human societies together.



Emotions help us navigate the complex social landscape of our lives. [Image: Gwenaël Piase, <https://goo.gl/d4EDKS>, CC BY-NC-SA 2.0, <https://goo.gl/hSpkVI>]

Intrapersonal Functions of Emotion

Emotions Help us Act Quickly with Minimal Conscious Awareness

Emotions are rapid information-processing systems that help us act with minimal thinking (Tooby & Cosmides, 2008). Problems associated with birth, battle, death, and seduction have occurred throughout evolutionary history and emotions evolved to aid humans in adapting

to those problems rapidly and with minimal conscious cognitive intervention. If we did not have emotions, we could not make rapid decisions concerning whether to attack, defend, flee, care for others, reject food, or approach something useful, all of which were functionally adaptive in our evolutionary history and helped us to survive. For instance, drinking spoiled milk or eating rotten eggs has negative consequences for our welfare. The emotion of disgust, however, helps us immediately take action by not ingesting them in the first place or by vomiting them out. This response is adaptive because it aids, ultimately, in our survival and allows us to act immediately without much thinking. In some instances, taking the time to sit and rationally think about what to do, calculating cost-benefit ratios in one's mind, is a luxury that might cost one one's life. Emotions evolved so that we can act without that depth of thinking.

Emotions Prepare the Body for Immediate Action



The emotion of disgust serves to protect us from toxins and contamination, of the physical and moral variety. [Image: Runs with Scissors, <https://goo.gl/FQRxGa>, CC BY-NC 2.0, <https://goo.gl/tgFydH>]

Emotions prepare us for behavior. When triggered, emotions orchestrate systems such as perception, attention, inference, learning, memory, goal choice, motivational priorities, physiological reactions, motor behaviors, and behavioral decision making (Cosmides & Tooby, 2000; Tooby & Cosmides, 2008). Emotions simultaneously activate certain systems and deactivate others in order to prevent the chaos of competing systems operating at the same time, allowing for coordinated responses to environmental stimuli (Levenson, 1999). For instance, when we are afraid, our bodies shut down temporarily unneeded digestive processes, resulting in saliva reduction (a dry mouth); blood flows disproportionately to the lower half of the

body; the visual field expands; and air is breathed in, all preparing the body to flee. Emotions initiate a system of components that includes subjective experience, expressive behaviors, physiological reactions, action tendencies, and cognition, all for the purposes of specific actions; the term “emotion” is, in reality, a metaphor for these reactions.

One common misunderstanding many people have when thinking about emotions, however,

is the belief that emotions must always directly produce action. This is not true. Emotion certainly *prepares* the body for action; but whether people actually engage in action is dependent on many factors, such as the context within which the emotion has occurred, the target of the emotion, the perceived consequences of one's actions, previous experiences, and so forth (Baumeister, Vohs, DeWall, & Zhang, 2007; Matsumoto & Wilson, 2008). Thus, emotions are just one of many determinants of behavior, albeit an important one.

Emotions Influence Thoughts

Emotions are also connected to thoughts and memories. Memories are not just facts that are encoded in our brains; they are colored with the emotions felt at those times the facts occurred (Wang & Ross, 2007). Thus, emotions serve as the neural glue that connects those disparate facts in our minds. That is why it is easier to remember happy thoughts when happy, and angry times when angry. Emotions serve as the affective basis of many attitudes, values, and beliefs that we have about the world and the people around us; without emotions those attitudes, values, and beliefs would be just statements without meaning, and emotions give those statements meaning. Emotions influence our thinking processes, sometimes in constructive ways, sometimes not. It is difficult to think critically and clearly when we feel intense emotions, but easier when we are not overwhelmed with emotions (Matsumoto, Hirayama, & LeRoux, 2006).

Emotions Motivate Future Behaviors

Because emotions prepare our bodies for immediate action, influence thoughts, and can be felt, they are important motivators of future behavior. Many of us strive to experience the feelings of satisfaction, joy, pride, or triumph in our accomplishments and achievements. At the same time, we also work very hard to avoid strong negative feelings; for example, once we have felt the emotion of disgust when drinking the spoiled milk, we generally work very hard to avoid having those feelings again (e.g., checking the expiration date on the label before buying the milk, smelling the milk before drinking it, watching if the milk curdles in one's coffee before drinking it). Emotions, therefore, not only influence immediate actions but also serve as an important motivational basis for future behaviors.

Interpersonal Functions of Emotion

Emotions are expressed both verbally through words and nonverbally through facial expressions, voices, gestures, body postures, and movements. We are constantly expressing

emotions when interacting with others, and others can reliably judge those emotional expressions (Elfenbein & Ambady, 2002; Matsumoto, 2001); thus, emotions have signal value to others and influence others and our social interactions. Emotions and their expressions communicate information to others about our feelings, intentions, relationship with the target of the emotions, and the environment. Because emotions have this communicative signal value, they help solve social problems by evoking responses from others, by signaling the nature of interpersonal relationships, and by providing incentives for desired social behavior (Keltner, 2003).



Emotions can act as signals to our friends and partners, conveying information about the quality of the relationship. [Image: mynameisharsha, <https://goo.gl/HY2XgV>, CC BY-SA 2.0, <https://goo.gl/rxiUsF>]

Emotional Expressions Facilitate Specific Behaviors in Perceivers

Because facial expressions of emotion are universal social signals, they contain meaning not only about the expressor's psychological state but also about that person's intent and subsequent behavior. This information affects what the perceiver is likely to do. People observing fearful faces, for instance, are more likely to produce approach-related behaviors, whereas people who observe angry faces are more likely to produce avoidance-related behaviors (Marsh, Ambady, & Kleck, 2005). Even subliminal presentation of smiles produces increases in how much beverage people pour and consume and how much they are willing to pay for it; presentation of angry faces decreases these behaviors (Winkielman, Berridge, & Wilbarger, 2005). Also, emotional displays evoke specific, complementary emotional responses from observers; for example, anger evokes fear in others (Dimberg & Ohman, 1996; Esteves, Dimberg, & Ohman, 1994), whereas distress evokes sympathy and aid (Eisenberg et al., 1989).

Emotional Expressions Signal the Nature of Interpersonal Relationships

Emotional expressions provide information about the nature of the relationships among

interactants. Some of the most important and provocative set of findings in this area come from studies involving married couples (Gottman & Levenson, 1992; Gottman, Levenson, & Woodin, 2001). In this research, married couples visited a laboratory after having not seen each other for 24 hours, and then engaged in intimate conversations about daily events or issues of conflict. Discrete expressions of contempt, especially by the men, and disgust, especially by the women, predicted later marital dissatisfaction and even divorce.

Emotional Expressions Provide Incentives for Desired Social Behavior

Facial expressions of emotion are important regulators of social interaction. In the developmental literature, this concept has been investigated under the concept of social referencing (Klinnert, Campos, & Sorce, 1983); that is, the process whereby infants seek out information from others to clarify a situation and then use that information to act. To date, the strongest demonstration of social referencing comes from work on the visual cliff. In the first study to investigate this concept, Campos and colleagues (Sorce, Emde, Campos, & Klinnert, 1985) placed mothers on the far end of the “cliff” from the infant. Mothers first smiled to the infants and placed a toy on top the safety glass to attract them; infants invariably began crawling to their mothers. When the infants were in the center of the table, however, the mother then posed an expression of fear, sadness, anger, interest, or joy. The results were clearly different for the different faces; no infant crossed the table when the mother showed fear; only 6% did when the mother posed anger, 33% crossed when the mother posed sadness, and approximately 75% of the infants crossed when the mother posed joy or interest.

Other studies provide similar support for facial expressions as regulators of social interaction. In one study (Bradshaw, 1986), experimenters posed facial expressions of neutral, anger, or disgust toward babies as they moved toward an object and measured the amount of inhibition the babies showed in touching the object. The results for 10- and 15-month olds were the same: anger produced the greatest inhibition, followed by disgust, with neutral the least. This study was later replicated (Hertenstein & Campos, 2004) using joy and disgust expressions, altering the method so that the infants were not allowed to touch the toy (compared with a distractor object) until one hour after exposure to the expression. At 14 months of age, significantly more infants touched the toy when they saw joyful expressions, but fewer touched the toy when the infants saw disgust.

Social and Cultural Functions of Emotion



Although there are cultural differences in the display of emotion, almost all infants start showing emotion such as smiling or reacting to their caretaker as early as 6 weeks after their birth. [Image: vgm8383, <https://goo.gl/jgfrDN>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

If you stop to think about many things we take for granted in our daily lives, we cannot help but come to the conclusion that modern human life is a colorful tapestry of many groups and individual lives woven together in a complex yet functional way. For example, when you're hungry, you might go to the local grocery store and buy some food. Ever stop to think about how you're able to do that? You might buy a banana that was grown in a field in southeast Asia being raised by farmers there, where they planted the tree, cared for it, and picked the fruit. They probably handed that fruit off to a distribution chain that allowed multiple people somewhere to use tools such as cranes, trucks, cargo bins, ships or airplanes (that were also created by multiple people somewhere) to bring that

banana to your store. The store had people to care for that banana until you came and got it and to barter with you for it (with your money). You may have gotten to the store riding a vehicle that was produced somewhere else in the world by others, and you were probably wearing clothes produced by some other people somewhere else.

Thus, human social life is complex. Individuals are members of multiple groups, with multiple social roles, norms, and expectations, and people move rapidly in and out of the multiple groups of which they are members. Moreover, much of human social life is unique because it revolves around cities, where many people of disparate backgrounds come together. This creates the enormous potential for social chaos, which can easily occur if individuals are not coordinated well and relationships not organized systematically.

One of the important functions of culture is to provide this necessary coordination and organization. Doing so allows individuals and groups to negotiate the social complexity of human social life, thereby maintaining social order and preventing social chaos. Culture does this by providing a meaning and information system to its members, which is shared by a group and transmitted across generations, that allows the group to meet basic needs of survival, pursue happiness and well-being, and derive meaning from life (Matsumoto & Juang, 2013). Culture is what allowed the banana from southeast Asia to appear on your table.

The Role of Emotions in the Function of Culture

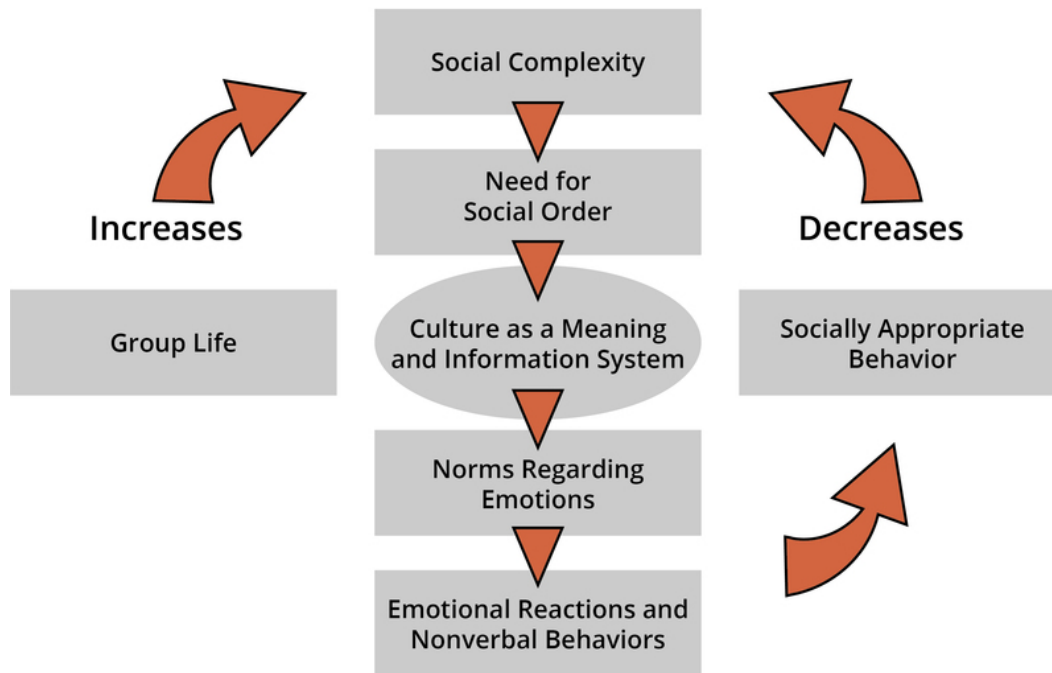


Figure 1: The Role of Emotions in the Function of Culture

Cultural transmission of the meaning and information system to its members is, therefore, a crucial aspect of culture. One of the ways this transmission occurs is through the development of worldviews (including attitudes, values, beliefs, and norms) related to emotions (Matsumoto & Hwang, 2013; Matsumoto et al., 2008). Worldviews related to emotions provide guidelines for desirable emotions that facilitate norms for regulating individual behaviors and interpersonal relationships. Our cultural backgrounds tell us which emotions are ideal to have, and which are not (Tsai, Knutson, & Fung, 2006). The cultural transmission of information related to emotions occurs in many ways, from childrearsers to children, as well as from the cultural products available in our world, such as books, movies, ads, and the like (Schönpflug, 2009; Tsai, Louie, Chen, & Uchida, 2007).

Cultures also inform us about what to do with our emotions—that is, how to manage or modify them—when we experience them. One of the ways in which this is done is through the management of our emotional expressions through cultural display rules (Friesen, 1972). These are rules that are learned early in life that specify the management and modification of our emotional expressions according to social circumstances. Thus, we learn that “big boys don’t cry” or to laugh at the boss’s jokes even though they’re not funny. By affecting how individuals express their emotions, culture also influences how people experience them as

well.

Because one of the major functions of culture is to maintain social order in order to ensure group efficiency and thus survival, cultures create worldviews, rules, guidelines, and norms concerning emotions because emotions have important intra- and interpersonal functions, as described above, and are important motivators of behavior. Norms concerning emotion and its regulation in all cultures serve the purpose of maintaining social order. Cultural worldviews and norms help us manage and modify our emotional reactions (and thus behaviors) by helping us to have certain kinds of emotional experiences in the first place and by managing our reactions and subsequent behaviors once we have them. By doing so,

our culturally moderated emotions can help us engage in socially appropriate behaviors, as defined by our cultures, and thus reduce social complexity and increase social order, avoiding social chaos. All of this allows us to live relatively harmonious and constructive lives in groups. If cultural worldviews and norms about emotions did not exist, people would just run amok having all kinds of emotional experiences, expressing their emotions and then behaving in all sorts of unpredictable and potentially harmful ways. If that were the case, it would be very difficult for groups and societies to function effectively, and even for humans to survive as a species, if emotions were not regulated in culturally defined ways for the common, social good. Thus, emotions play a critical role in the successful functioning of any society and culture.



Cultural display rules teach us how to manage our emotions. For example, in many Asian countries children are taught to mute their emotions, especially negative emotions like anger. [Image: john.gillespie, <https://goo.gl/gTdPYb>, CC BY-SA 3.0, <https://goo.gl/eLCn2O>]

Outside Resources

Alberta, G. M., Rieckmann, T. R., & Rush, J. D. (2000). Issues and recommendations for teaching an ethnic/culture-based course. *Teaching of Psychology*, 27, 102-107. doi:10.1207/S15328023TOP2702_05

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NPR News: Science Of Sadness And Joy: 'Inside Out' Gets Childhood Emotions Right

<http://www.npr.org/sections/health-shots/2015/06/13/413980258/science-of-sadness-and-joy-inside-out-gets-childhood-emotions-right>

Online Psychology Laboratory: Motivation and Emotion resources

<http://opl.apa.org/Resources.aspx#Motivation>

Web: See how well you can read other people's facial expressions of emotion

<http://www.humintell.com/free-demos/>

Discussion Questions

1. When emotions occur, why do they simultaneously activate certain physiological and psychological systems in the body and deactivate others?

2. Why is it difficult for people to act rationally and think happy thoughts when they are angry? Conversely, why is it difficult to remember sad memories or have sad thoughts when people are happy?
3. You're walking down a deserted street when you come across a stranger who looks scared. What would you say? What would you do? Why?
4. You're walking down a deserted street when you come across a stranger who looks angry. What would you say? What would you do? Why?
5. Think about the messages children receive from their environment (such as from parents, mass media, the Internet, Hollywood movies, billboards, and storybooks). In what ways do these messages influence the kinds of emotions that children should and should not feel?

Vocabulary

Cultural display rules

These are rules that are learned early in life that specify the management and modification of emotional expressions according to social circumstances. Cultural display rules can work in a number of different ways. For example, they can require individuals to express emotions “as is” (i.e., as they feel them), to exaggerate their expressions to show more than what is actually felt, to tone down their expressions to show less than what is actually felt, to conceal their feelings by expressing something else, or to show nothing at all.

Interpersonal

This refers to the relationship or interaction between two or more individuals in a group. Thus, the interpersonal functions of emotion refer to the effects of one’s emotion on others, or to the relationship between oneself and others.

Intrapersonal

This refers to what occurs within oneself. Thus, the intrapersonal functions of emotion refer to the effects of emotion to individuals that occur physically inside their bodies and psychologically inside their minds.

Social and cultural

Society refers to a system of relationships between individuals and groups of individuals; culture refers to the meaning and information afforded to that system that is transmitted across generations. Thus, the social and cultural functions of emotion refer to the effects that emotions have on the functioning and maintenance of societies and cultures.

Social referencing

This refers to the process whereby individuals look for information from others to clarify a situation, and then use that information to act. Thus, individuals will often use the emotional expressions of others as a source of information to make decisions about their own behavior.

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Emotion Experience and Well-Being

Brett Ford & Iris B. Mauss

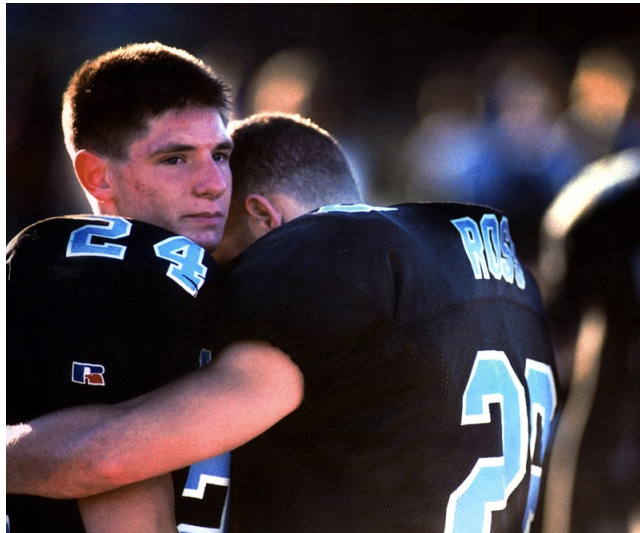
Emotions don't just feel good or bad, they also contribute crucially to people's well-being and health. In general, experiencing positive emotions is good for us, whereas experiencing negative emotions is bad for us. However, recent research on emotions and well-being suggests this simple conclusion is incomplete and sometimes even wrong. Taking a closer look at this research, the present module provides a more complex relationship between emotion and well-being. At least three aspects of the emotional experience appear to affect how a given emotion is linked with well-being: the intensity of the emotion experienced, the fluctuation of the emotion experienced, and the context in which the emotion is experienced. While it is generally good to experience more positive emotion and less negative emotion, this is not always the guide to the good life.

Learning Objectives

- Describe the general pattern of associations between emotion experience and well-being.
- Identify at least three aspects of emotion experience beyond positivity and negativity of the emotion that affect the link between emotion experience and well-being.

How we feel adds much of the flavor to life's highest—and lowest—moments. Can you think of an important moment in your life that didn't involve strong feelings? In fact, it might be hard to recall *any* times when you had no feeling at all. Given how saturated human life is with feelings, and given how profoundly feelings affect us, it is not surprising that much theorizing and research has been devoted to uncovering how we can optimize our feelings, or, "emotion experiences," as they are referred to in psychological research.

Feelings contribute to well-being



Although we tend to think we should always strive for “positive” or “feel-good” emotions, in some situations a negative emotion might be best. For example feeling disappointment when experiencing a failure might help motivate you for future success.

[Image: CC0 Public Domain, <https://goo.gl/m25gce>]

it may be helpful to think of “positive” and “negative” as descriptive terms used to discuss two different types of experiences, rather than a true value judgment. Thus, whether positive or negative emotions are good or bad for us is an empirical question.

So, which emotions are the “best” ones to feel? Take a moment to think about how you might answer this question. At first glance, the answer might seem obvious. Of course, we should experience as much positive emotion and as little negative emotion as possible! Why? Because it is pleasant to experience positive emotions and it is unpleasant to experience negative emotions (Russell & Barrett, 1999). The conclusion that positive feelings are good and negative feelings are bad might seem so obvious as not to even warrant the question, much less bona fide psychological research. In fact, the very labels of “positive” and “negative” imply the answer to this question. However, for the purposes of this module,

As it turns out, this empirical question has been on the minds of theorists and researchers for many years. Such psychologists as Alice Isen, Charles Carver, Michael Scheier, and, more recently, Barbara Fredrickson, Dacher Keltner, Sonja Lyubomirsky, and others began asking whether the effects of feelings could go beyond the obvious momentary pleasure or displeasure. In other words, can emotions do more for us than simply make us feel good or bad? This is not necessarily a new question; variants of it have appeared in the texts of thinkers such as Charles Darwin (1872) and Aristotle (1999). However, modern psychological research has provided empirical evidence that feelings are not just inconsequential byproducts. Rather, each emotion experience, however fleeting, has effects on cognition, behavior, and the people around us. For example, feeling happy is not only pleasant, but is also useful to feel when in social situations because it helps us be friendly and collaborative, thus promoting our positive relationships. Over time, the argument goes, these effects add up to have tangible effects on people’s well-being (good mental and physical health).

A variety of research has been inspired by the notion that our emotions are involved in, and maybe even causally contribute to, our well-being. This research has shown that people who experience more frequent positive emotions and less frequent negative emotions have higher well-being (e.g., Fredrickson, 1998; Lyubomirsky, King, & Diener, 2005), including increased life satisfaction (Diener, Sandvik, & Pavot, 1991), increased physical health (Tugade, Fredrickson, & Barrett, 2004; Veenhoven, 2008), greater resilience to stress (Folkman & Moskowitz, 2000; Tugade & Fredrickson, 2004), better social connection with others (Fredrickson, 1998), and even longer lives (Veenhoven, 2008). Notably, the effect of positive emotion on longevity is about as powerful as the effect of smoking! Perhaps most importantly, some research directly supports that emotional experiences *cause* these various outcomes rather than being just a consequence of them (Fredrickson, Cohn, Coffey, Pek, & Finkel, 2008; Lyubomirsky et al., 2005).

At this point, you might be tempted to conclude that you should always strive to experience as much positive emotion and as little negative emotion as possible. However, recent research suggests that this conclusion may be premature. This is because this conclusion neglects three central aspects of the emotion experience. First, it neglects the intensity of the emotion: Positive and negative emotions might not have the same effect on well-being at all intensities. Second, it neglects how emotions fluctuate over time: Stable emotion experiences might have quite different effects from experiences that change a lot. Third, it neglects the context in which the emotion is experienced: The context in which we experience an emotion might profoundly affect whether the emotion is good or bad for us. So, to address the question “Which emotions should we feel?” we must answer, “It depends!” We next consider each of the three aspects of feelings, and how they influence the link between feelings and well-being.



Not only do the emotions we feel vary by the context, but also the emotions we should feel depend on the circumstances, too. [Image: puppywind, <https://goo.gl/BQKhKK>, CC BY-NC-ND 2.0, <https://goo.gl/h9gK3o>]

The intensity of the emotion matters

Experiencing more *frequent* positive emotions is generally beneficial. But does this mean that we should strive to feel as *intense* positive emotion as possible? Recent research suggests that this unqualified conclusion might be wrong.

In fact, experiencing very high levels of positive emotion may be harmful (Gruber, 2011; Oishi, Diener, & Lucas, 2007). For instance, experiencing *very* high levels of positive emotion makes individuals more likely to engage in risky behaviors, such as binge eating and drug use (Cyders & Smith, 2008; Martin et al., 2002). Furthermore, intense positive emotion is associated with the experience of mania (Gruber et al., 2009; Johnson, 2005). It appears that the experience of positive emotions follows an inverted U-shaped curve in relation to well-being: more positive emotion is linked with increased well-being, but only up to a point, after which even more positive emotion is linked with decreased well-being (Grant & Schwartz, 2011). These empirical findings underscore the sentiment put forth long ago by the philosopher Aristotle: Moderation is key to leading a good life (1999).

Too much positive emotion may pose a problem for well-being. Might too little negative emotion similarly be cause for concern? Although there is limited empirical research on this subject, initial research suggests supports this idea. For example, people who aim *not* to feel negative emotion are at risk for worse well-being and adaptive functioning, including lower life satisfaction, lower social support, worse college grades, and feelings of worse physical health (Tamir & Ford, 2012a). Similarly, feeling too little embarrassment in response to a social faux pas may damage someone's social connections if they aren't motivated by their embarrassment to make amends (Keltner & Buswell, 1997). Low levels of negative emotion also seem to be involved in some forms of psychopathology. For instance, blunted sadness in response to a sad situation is a characteristic of major depressive disorder (Rottenberg, Gross, & Gotlib, 2005) and feeling too little fear is a hallmark of psychopathy (Marsh et al., 2008; Patrick, 1994).

In sum, this first section suggests that the conclusion "Of course we should experience as much positive emotions and as little negative emotions as possible" is sometimes wrong. As it turns out, there can be too much of a good thing and too little of a bad thing.

The fluctuation of the emotion matters

Emotions naturally vary—or fluctuate—over time (Davidson, 1998). We probably all know someone whose emotions seem to fly everywhere—one minute they're ecstatic, the next they're upset. We might also know a person who is pretty even-keeled, moderately happy, with only modest fluctuations across time. When looking only at *average* emotion experience,

say across a month, both of these people might appear identical: moderately happy. However, underlying these identical averages are two very different patterns of fluctuation across time. Might these **emotion fluctuations** across time—beyond average intensity—have implications for well-being?

Overall, the available research suggests that how much emotions fluctuate does indeed matter. In general, greater fluctuations are associated with worse well-being. For example, higher fluctuation of positive emotions—measured either within a single day or across two weeks—was linked with lower well-being and greater depression (Gruber, Kogan, Quoidbach, & Mauss, 2013). Fluctuation in negative emotions, in turn, has been linked with increased depressive symptoms (Peeters, Berkhof, Delespaul, Rottenberg, & Nicolson, 2003), borderline personality disorder (Trull et al., 2008), and neuroticism (Eid & Diener, 1999). These associations tend to hold even when controlling for average levels of positive or negative emotion, which means that beyond the overall intensity of positive or negative emotion, the fluctuation of one's emotions across time is associated with well-being. While it is not entirely clear why fluctuations are linked to worse well-being, one explanation is that strong fluctuations are indicative of emotional instability (Kuppens, Oravecz, & Tuerlinckx, 2010).

Of course, this should not be taken to mean that we should rigidly feel the exact same way every minute of every day, regardless of context. After all, psychological flexibility—or the ability to adapt to changing situational demands and experience emotions accordingly—has generally demonstrated beneficial links with well-being (Bonanno, Papa, Lalande, Westphal, & Coifman, 2004; Kashdan, & Rottenberg, 2010). The question remains, however: what exact amount of emotional fluctuation constitutes unhealthy instability and what amount of emotional fluctuation constitutes healthy flexibility.

Again, then, we must qualify the conclusion that it is always better to experience more positive emotions and less negative emotions. The degree to which emotions fluctuate across time plays an important role. Overall, relative stability (but not rigidity) in emotion experience appears to be optimal for well-being.



Throughout our lives, our emotions vary such that we experience great highs and great lows. However, if you average those emotions over time, we tend to end up somewhere in the middle. [Image: Moazzam Brohi, <https://goo.gl/acZniv>, CC BY 2.0, <https://goo.gl/BRvSA7>]

The context of the emotion experience matters

This module has already discussed two features of emotion experiences that affect how they relate to well-being: the intensity of the emotion and the fluctuation of the emotion over time. However, neither of these features takes into account the context in which the emotion is experienced. At least three different contexts may critically affect the links between emotion and well-being: (1) the external environment in which the emotion is being experienced, (2) the other emotional responses (e.g., physiology, facial behavior) that are currently activated, and (3) the other emotions that are currently being experienced.

The external environment



Feeling an emotion that matches the person's context (e.g., experiencing happiness at a birthday party) is typically the most functional or beneficial emotion to feel. [Image: OakleyOriginals, <https://goo.gl/lxflsq>, CC BY 2.0, <https://goo.gl/BRvSA7>]

Emotions don't occur within a vacuum. Instead, they are usually elicited by and experienced within specific situations that come in many shapes and sizes—from birthday parties to funerals, job interviews to mundane movie nights. The situation in which an emotion is experienced has strong implications for whether a given emotion is the “best” emotion to feel. Take happiness, for example. Feeling happiness at a birthday party may be a great idea. However, having the exact same experience of happiness at a funeral would likely not bode well for your well-being.

When considering how the environment influences the link between emotion and well-being, it is important to understand that each emotion has its own function. For example, although fear is a negative emotion, fear helps us notice and avoid threats to our safety (Öhman & Mineka, 2001), and may thus be the “best” emotion to feel in dangerous situations. Happiness can help people cooperate with others, and may thus be the best emotion to feel when we need to collaborate (e.g., Van Kleef, van Dijk, Steinel, & van Beest, 2008). Anger can energize people to compete or fight with others, and may thus be advantageous to experience it in confrontations (e.g., Tamir & Ford, 2012b; Van Kleef et al., 2008). It might be disadvantageous to experience

happiness (a positive emotion) when we need to fight with someone; in this situation, it might be better to experience anger (a negative emotion). This suggests that emotions' implications for well-being are not determined only by whether they are positive or negative but also by whether they are well-matched to their context.

In support of this general idea, people who experience emotions that fit the context at hand are more likely to recover from depression and trauma (Bonanno et al., 2004; Rottenberg, Kasch, Gross, & Gotlib, 2002). Research has also found that participants who want to feel emotions that match the context at hand (e.g., anger when confronting someone)—even if that emotion was negative—are more likely to experience greater well-being (Tamir & Ford, 2012a). Conversely, people who pursue emotions without regard to context—even if those emotions are positive, like happiness—are more likely to experience lower subjective well-being, more depression, greater loneliness, and even worse grades (Ford & Tamir, 2012; Mauss et al., 2012; Mauss, Tamir, Anderson, & Savino, 2011; Tamir & Ford, 2012a).

In sum, this research demonstrates that regardless of whether an emotion is positive or negative, the context in which it is experienced critically influences whether the emotion helps or hinders well-being.

Other emotional responses



If we experience the emotion of amusement (from seeing something funny), we often have the physiological response to laugh. This is an example of emotion coherence, where we express a particular behavior associated with a particular emotion. [Image: Ed Schipul, <https://goo.gl/7NUYmR>, CC BY-SA 2.0, <https://goo.gl/rxiUsF>]

The subjective experience of an emotion—what an emotion *feels* like—is only one aspect of an emotion. Other aspects include behaviors, facial expressions, and physiological activation (Levenson, 1992). For example, if you feel excited about having made a new friend, you might want to be near that person, you might smile, and your heart might be beating faster as you do so. Often, these different responses travel together, meaning that when we feel an emotion we *typically* have corresponding behaviors and physiological responses (e. g., Ekman, 1972; Levenson, 1992). The degree to which responses travel together has sometimes been referred to as emotion coherence (Mauss, Levenson,

McCarter, Wilhelm, & Gross, 2005). However, these different responses do not co-occur in all instances and for all people (Bradley & Lang, 2000; Mauss et al., 2005; for review, see Fridlund, Ekman, & Oster, 1987). For example, some people may choose not to express an emotion they are feeling internally (English & John, 2013), which would result in lower coherence.

Does coherence—above and beyond emotion experience per se—matter for people's well-being? To examine this question, one study measured participants' emotion coherence by showing them a funny film clip of stand-up comedy while recording their experience of positive emotion as well as their behavioral displays of positive emotion (Mauss, Shallcross, et al., 2011). As predicted, participants differed quite a bit in their coherence. Some showed almost perfect coherence between their behavior and experience, whereas others' behavior and experience corresponded not much at all. Interestingly, the more that participants' behavior and experience cohered in the laboratory session, the lower levels of depressive symptoms and the higher levels of well-being they experienced 6 months later. This effect was found when statistically controlling for overall intensity of positive emotions experienced. In other words, experiencing high levels of positive emotion aided well-being *only if it was accompanied by corresponding positive facial expressions*.

But *why* would coherence of different emotional responses predict well-being? One of the key functions of an emotion is social communication (Keltner & Haidt, 1999), and arguably, successful social communication depends on whether an individual's emotions are being *accurately* communicated to others. When someone's emotional behavior doesn't match their experience it may disrupt communication because it could make the individual appear confusing or inauthentic to others. In support of this theory, the above study found that lower coherence was associated with worse well-being because people with lower coherence felt less socially connected to others (Mauss, Shallcross, et al., 2011). These findings are also consistent with a large body of research examining the extent to which people mask the outward display of an emotional experience, or *suppression*. This research has demonstrated that people who habitually use suppression not only experience worse well being (Gross & John, 2003), but they also seem to be particularly worse off with regard to their social relationships (Srivastava, Tamir, McGonigal, John, & Gross, 2009).

These findings underscore the importance of examining whether an individual's experience is traveling together with his or her emotional responses, above and beyond overall levels of subjective experience. Thus, to understand how emotion experiences predict well-being, it is important not only to consider the experience of an emotion, but also the other emotional responses currently activated.

Other emotions

Up until now, we have treated emotional experiences as though people can only experience one emotion at a time. However, it should be kept in mind that positive and negative emotions are not simply the opposite of one another. Instead, they tend to be independent of one another, which means that a person can feel positive and negative emotions at the same time (Larsen, McGraw, Mellers, & Cacioppo, 2004). For example, how does it feel to win a prize when you expected a greater prize? Given “what might have been,” situations like this can elicit both happiness and sadness. Or, take “schadenfreude” (a German term for deriving pleasure from someone else’s misfortune), or “aviman” (an Indian term for prideful, loving anger), or nostalgia (an English term for affectionate sadness about something from the past): these terms capture the notion that people can feel both positively and negatively within the same emotional experience. And as it turns out, the other emotions that someone feels (e.g., sadness) during the experience of an emotion (e.g., happiness) influence whether that emotion experience has a positive or negative effect on well-being.

Importantly, the extent to which someone experiences different emotions at the same time—or *mixed emotions*—may be beneficial for their well-being. Early support for this theory was provided by a study of bereaved spouses. In the study, participants were asked to talk about their recently deceased spouse, which undoubtedly elicited strong negative emotions. However, some participants expressed positive emotions in addition to the negative ones, and it was those participants who recovered more quickly from their loss (Bonanno & Keltner, 1997). A recent study provides additional support for the benefits of mixed emotions, finding that adults who experienced more mixed emotions over a span of 10 years were physically healthier than adults whose experience of mixed emotions did not increase over time (Hershfield, Scheibe, Sims & Carstensen, 2013). Indeed, individuals who can experience positive emotions even in the face of negative emotions are more likely to cope successfully with stressful situations (Larsen, Hemenover, Norris, & Cacioppo, 2003).



What experiences can you recall where you felt mixed emotions? Happiness and disappointment? Hope and fear? Admiration and envy? [Image: Ron Cogswell, <https://goo.gl/JKuzmU>, CC BY 2.0, <https://goo.gl/BRvSA7>]

Why would mixed emotions be beneficial for well-being? Stressful situations often elicit negative emotions, and recall that negative emotions have some benefits, as we outlined

above. However, so do positive emotions, and thus having the ability to “take the good with the bad” might be another key component of well-being. Again, experiencing more positive emotion and less negative emotion may not always be optimal. Sometimes, a combination of both may be best.

Conclusion

Are emotions just fleeting experiences with no consequence beyond our momentary comfort or discomfort? A variety of research answers a firm “no”—emotions are integral predictors of our well-being. This module examined how, exactly, emotion experience might be linked to well-being. The obvious answer to this question is: *of course*, experiencing as much positive emotions and as little negative emotions as possible is good for us. But although this is true in general, recent research suggests that this obvious answer is incomplete and sometimes even wrong. As philosopher Robert Solomon said, “Living well is not just maximizing the good feelings and minimizing the bad. (...) A happy life is not necessarily filled with happy moments” (2007, p. 86).

Outside Resources

Journal: If you are interested in direct access to research on emotion, take a look at the journal *Emotion*

<http://www.apa.org/pubs/journals/emo/index.aspx>

Video: Check out videos of expert emotion researchers discussing their work

http://www.youtube.com/playlist?list=PLh9mgdi4rNew731mjlZn43G_Y5otqKzJA

Video: See psychologist Daniel Gilbert and other experts discussing current research on emotion in the PBS series *This Emotional Life*

<http://video.pbs.org/program/this-emotional-life/>

Discussion Questions

1. Much research confirms the relative benefits of positive emotions and relative costs of negative emotions. Could positive emotions be detrimental, or could negative emotions be beneficial? Why or why not?
2. We described some contexts that influence the effects of emotional experiences on well-being. What other contexts might influence the links between emotions and well-being? Age? Gender? Culture? How so?
3. How could you design an experiment that tests...(A) When and why it is beneficial to feel a negative emotion such as sadness? (B) How is the coherence of emotion behavior and emotion experience linked to well-being? (C) How likely a person is to feel mixed (as compared to simple) emotions?

Vocabulary

Emotion

An experiential, physiological, and behavioral response to a personally meaningful stimulus.

Emotion coherence

The degree to which emotional responses (subjective experience, behavior, physiology, etc.) converge with one another.

Emotion fluctuation

The degree to which emotions vary or change in intensity over time.

Well-being

The experience of mental and physical health and the absence of disorder.

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Emotional Intelligence

Marc Brackett, Sarah Delaney & Peter Salovey

In this module, we review the construct of emotional intelligence by examining its underlying theoretical model, measurement tools, validity, and applications in real-world settings. We use empirical research from the past few decades to support and discuss competing definitions of emotional intelligence and possible future directions for the field.

Learning Objectives

- Understand the theoretical foundations of emotional intelligence and the relationship between emotion and cognition.
- Distinguish between mixed and ability models of emotional intelligence.
- Understand various methods for measuring emotional intelligence.
- Describe emotional intelligence's evolution as a theoretical, success-oriented, and achievement-based framework.
- Identify and define key concepts of emotional intelligence (including emotion regulation, expression of emotion, understanding emotion, etc.) and the ways they contribute to decision making, relationship building, and overall well-being.

Introduction

Imagine you are waiting in line to buy tickets to see your favorite band. Knowing tickets are limited and prices will rise quickly, you showed up 4 hours early. Unfortunately, so did everyone else. The line stretches for blocks and hasn't moved since you arrived. It starts to rain. You

are now close to Will Call when you notice three people jump ahead of you to join their friends, who appear to have been saving a spot for them. They talk loudly on their cellphones as you inch forward, following the slow procession of others waiting in line. You finally reach the ticket counter only to have the clerk tell you the show is sold out. You notice the loud group off to the side, waving their tickets in the air. At this exact moment, a fiery line of emotion shoots through your whole body. Your heart begins to race, and you feel the urge to either slam your hands on the counter or scream in the face of those you believe have slighted you. What are these feelings, and what will you do with them?



After a serious disappointment or injustice how hard is it to keep control of your emotions? [Image: DCist, <https://goo.gl/o9EZOG>, CC BY-2.0, <https://goo.gl/zHmGV2>]

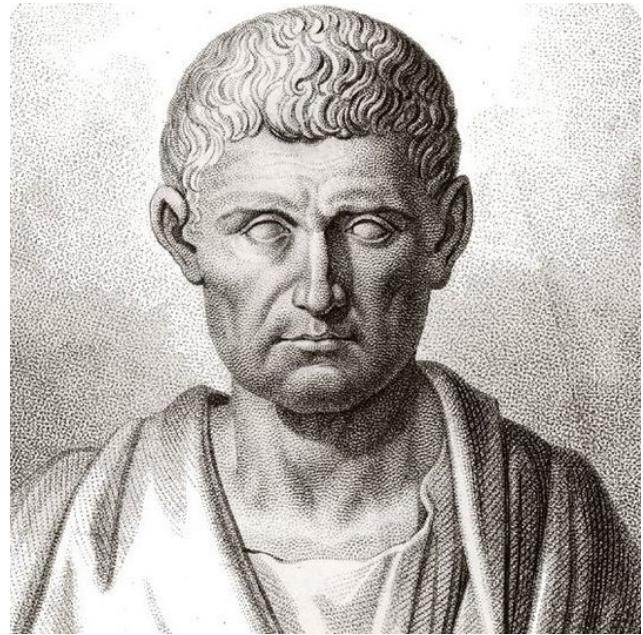
Emotional intelligence (EI) involves the idea that cognition and emotion are interrelated. From this notion stems the belief that emotions influence decision making, relationship building, and everyday behavior. After spending hours waiting eagerly in the pouring rain and having nothing to show for it, is it even possible to squelch such intense feelings of anger due to injustice? From an EI perspective, emotions are active mental processes that can be managed, so long as individuals develop the knowledge and skills to do so. But how, exactly, do we reason with our emotions? In other words, how intelligent is our emotion system?

To begin, we'll briefly review the concept of standard, or general, intelligence. The late American psychologist, David Wechsler, claimed that intelligence is the "global capacity of an individual to think rationally, act purposefully, and deal effectively with their environment" (Wechsler, 1944). If we choose to accept this definition, then intelligence is an operational process through which we learn to utilize our internal abilities in order to better navigate our surroundings—a process that is most certainly similar to, if not impacted by, our emotions. In 1990, Drs. Peter Salovey and John D. Mayer first explored and defined EI. They explained EI as "the ability to monitor one's own and others' feelings and emotions, to discriminate among them and use this information to guide one's thinking and actions" (Salovey & Mayer, 1990). EI, according to these researchers, asserts that all individuals possess the ability to leverage their emotions to enhance thinking, judgment, and behavior. This module aims to unpack this theory by exploring the growing

empirical research on EI, as well as what can be learned about its impact on our daily lives.

History of EI

Traditionally, many psychologists and philosophers viewed cognition and emotion as separate domains, with emotion posing a threat to productive and rational thinking. Have you ever been told not to let your emotions get in the way of your decisions? This separation of passion and reason stretches as far back as early ancient Greece (Lyons, 1999). Additionally, mid-20th century scholars explained emotions as mentally destabilizing forces (Young, 1943). Yet, there are traces throughout history where the intersection of emotion and cognition has been theoretically questioned. In 350 B.C.E., the famous Greek philosopher Aristotle wrote, “some men . . . if they have first perceived and seen what is coming and have first roused themselves and their calculative



Perhaps Aristotle might have revised his statement about people not being “defeated by their emotion” if he was ever stuck in rush hour traffic. [CC0 Public Domain, <https://goo.gl/m25gce>]

faculty, are not defeated by their emotion, whether it be pleasant or painful”(Aristotle, trans. 2009, Book VII, Chapter 7, Section 8). Still, our social interactions and experiences suggest this belief has undergone centuries of disregard, both in Western and Eastern cultures. These are the same interactions that teach us to “toughen up” and keep our emotions hidden. So, how did we arrive at EI—a scientific theory that claims *all* individuals have access to a “calculative faculty” through emotion?

In the early 1970s, many scientists began to recognize the limitations of the Intelligence Quotient (IQ)—the standardized assessment of intelligence. In particular, they noticed its inability to explain differences among individuals unrelated to just cognitive ability alone. These frustrations led to the advancement of more inclusive theories of intelligence such as Gardner’s multiple intelligences theory (1983/1993) and Sternberg’s triarchic theory of intelligence (1985). Researchers also began to explore the influence of moods and emotions on thought processes, including judgment (Isen, Shalcker, Clark, & Karp, 1978) and memory (Bower, 1981). It was through these theoretical explorations and empirical studies that the

concept of EI began to take shape.

Today, the field of EI is extensive, encompassing varying perspectives and measurement tools. Some attribute this growth to Daniel Goleman's popularization of the construct in his 1995 book, *Emotional Intelligence: Why It Can Matter More Than IQ*. Generating public appeal, he focused on EI's connection to personal and professional success. Goleman's model of EI includes a blend of emotion-related skills, traditional cognitive intelligence, and distinct personality traits. This embellished conceptualization of EI, followed by an increase in EI literature, contributed, at least in part, to conflicting definitional and measurement models within the field.

Models and Measures of EI

Many researchers would agree that EI theory will only be as successful as its form of measurement. Today, there are three primary models of EI: the ability model (Mayer & Salovey 1997; Salovey & Mayer, 1990), mixed models (Bar-On, 2006; Boyatzis & Sala, 2004), and the trait EI model (Petrides & Furnham, 2003).

Ability models approach EI as a standard intelligence that utilizes a distinct set of mental abilities that (1) are intercorrelated, (2) relate to other extant intelligences, and (3) develop with age and experience (Mayer, Caruso, & Salovey, 1999; Mayer, Salovey, Caruso, & Sitarenios, 2003). In contrast, both mixed and trait models define and measure EI as a set of perceived abilities, skills, and personality traits.

Ability Models: Mayer and Salovey Four-Branch Model of EI

In this section, we describe the EI (**Four-Branch**) model espoused by Mayer and Salovey (1997). This model proposes that four fundamental emotion-related abilities comprise EI: (1) perception/expression of emotion, (2) use of emotion to facilitate thinking, (3) understanding of emotion, and (4) management of emotion in oneself and others.

1. Perception of Emotion

Perception of emotion refers to people's capacity to identify emotions in themselves and others using facial expressions, tone of voice, and body language (Brackett et al., 2013). Those skilled in the perception of emotion also are able to express emotion accordingly and communicate emotional needs. For example, let's return to our opening scenario. After being

turned away at the ticket booth, you slowly settle into the reality that you cannot attend the concert. A group of your classmates, however, managed to buy tickets and are discussing their plans at your lunch table. When they ask if you are excited for the opening band, you shrug and pick at your food. If your classmates are skilled at perception of emotion, then they will read your facial expression and body language and determine that you might be masking your true feelings of disappointment, frustration, or disengagement from the conversation. As a result, they might ask you if something is wrong or choose not to talk about the concert in your presence.

2. Use of Emotion to Facilitate Thinking

Using emotion to enhance cognitive activities and adapt to various situations is the second component of EI. People who are skilled in this area understand that some emotional states are more optimal for targeted outcomes than others. Feeling frustrated over the concert tickets may be a helpful mindset as you are about to play a football game or begin a wrestling match. The high levels of adrenaline associated with frustration may boost your energy and strength, helping you compete. These same emotions, however, will likely impede your ability to sit at your school desk and solve algebra problems or write an essay.

Individuals who have developed and practiced this area of EI actively generate emotions that support certain tasks or objectives. For example, a teacher skilled in this domain may recognize that her students need to experience positive emotions, like joy or excitement, in order to succeed when doing creative work such as brainstorming or collaborative art projects. She may plan accordingly by scheduling these activities for after recess, knowing students will likely come into the classroom cheerful and happy from playing outside. Making decisions based on the impact that emotional experiences may have on actions and behavior is an essential component of EI.



Which emotions would serve you best during a football game? Would the same emotions be useful for a chess match? Matching the emotion to the task at hand is a valuable skill to have. [Image: Ian Sane, <https://goo.gl/IKuqyz>, CC BY 2.0, <https://goo.gl/zHmGV2>]

3. Understanding of Emotion

EI also includes the ability to differentiate between emotional states, as well as their specific causes and trajectories. Feelings of sadness or disappointment can result from the loss of a person or object, such as your concert tickets. Standing in the rain, by most standards, is merely a slight annoyance. However, waiting in the rain for hours in a large crowd will likely result in irritation or frustration. Feeling like you have been treated unfairly when someone cuts in line and takes the tickets you feel you deserved can cause your unpleasantness to escalate into anger and resentment. People skilled in this area are aware of this emotional trajectory and also have a strong sense of how multiple emotions can work together to produce another. For instance, it is possible that you may feel contempt for the people who cut in front of you in line. However, this feeling of contempt does not arise from anger alone. Rather, it is the combination of anger and disgust by the fact that these individuals, unlike you, have disobeyed the rules. Successfully discriminating between negative emotions is an important skill related to understanding of emotion, and it may lead to more effective emotion management (Feldman Barret, Gross, Christensen, & Benvenuto, 2001).

4. Management of Emotion

Emotion management includes the ability to remain open to a wide range of emotions, recognize the value of feeling certain emotions in specific situations, and understand which short- and long-term strategies are most efficient for emotion regulation (Gross, 1998). Anger seems an appropriate response to falling short of a goal (concert tickets) that you pursued both fairly and patiently. In fact, you may even find it valuable to allow yourself the experience of this feeling. However, this feeling will certainly need to be managed in order to prevent aggressive, unwanted behavior. Coming up with strategies, such as taking a deep breath and waiting until you feel calm before letting the group ahead of you know they cut in line, will allow you to regulate your anger and prevent the situation from escalating. Using this strategy may even let you gain insight into other perspectives—perhaps you learn they had already purchased their tickets and were merely accompanying their friends.

Measuring EI with Performance Measures

While self-report tests are common in psychology, ability models of EI require a different approach: performance measures. Performance measures require respondents to demonstrate their four emotion skills (Mayer & Salovey, 1997) by solving emotion-related problems. Among these measures, the Mayer-Salovey-Caruso Emotional Intelligence Test

(MSCEIT) (Mayer, Salovey, & Caruso, 2002) is the most commonly used. The MSCEIT is a 141-item test comprised of a total of eight tasks, two per each of the four emotion abilities. To measure emotion management, for example, respondents are asked to read through scenarios involving emotionally charged conflicts and then asked to evaluate the effectiveness of different resolutions. For a comprehensive review of the MSCEIT and other performance-assessment tools, please see Rivers, Brackett, Salovey, and Mayer (2007).

Mixed and Trait Models of EI

Unlike ability models, **mixed models** offer a broad definition of EI that combines mental abilities with personality traits such as optimism, motivation, and stress tolerance (see Cherniss, 2010, for a review). The two most widely used mixed models are the Boyatzis-Goleman model (Boyatzis & Sala, 2004) and the Bar-On model of emotional-social intelligence (Bar-On, 2006). The Boyatzis-Goleman model divides EI competencies into four groups: self-awareness, self-management, social awareness, and relationship management. Similarly, the Bar-On model offers five main components of EI: intrapersonal skills, interpersonal skills, adaptability, stress management, and mood. Developers of the trait EI model (Petrides & Furnham, 2003) explain EI as a constellation of self-perceived, emotion-related personality traits.



As an alternative to somewhat unreliable self-report measures, researchers have been using fMRI studies to detect changes in brain function when a self-report type of question is given to a participant. [Image: NIH Image Gallery, <https://goo.gl/aWZBvI>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

Mixed and Trait Model Assessment: Self-Report

Self-report assessments—surveys that ask respondents to report their own emotional skills—are most often associated with mixed and trait models. Self-report measures are usually quick to administer. However, many researchers argue that their vulnerability to social-desirability biases and faking are problematic (Day & Carroll, 2008). In addition, there is wide speculation concerning the potential for inaccurate judgments of personal ability and skill on behalf of responders (e.g., Paulhus, Lysy, & Yik, 1998). Self-report measures have been shown to lack

discriminant validity from existing personality measures and have very low correlations with ability measures of EI (Brackett & Mayer, 2003; Brackett, Rivers, Shiffman, Lerner, & Salovey, 2006). According to Mayer and colleagues (2008), self-report tests may show reliability for individual personalities, but should not be considered EI because performance tests are the gold standard for measuring intelligence.

Although tensions between ability and mixed or trait model approaches appear to divide the field, competing definitions and measurements can only enhance the quality of research devoted to EI and its impact on real-world outcomes.

Room for Debate

While mixed and trait models shed some light on the concept of EI, many researchers feel these approaches undermine the EI construct as a discrete and measurable mental ability. EI, when conceptualized as an ability, most accurately describes the relationship between cognition and emotion by accounting for changes in individual outcomes that are often missed when focusing solely on cognitive intelligence or personality traits (O'Boyle, Humphrey, Pollack, Hawver, & Story, 2010). What's more, among adults, personality traits provide little room for malleability, making development in these areas difficult even when combined with emotional skills. For example, characteristics such as agreeableness and neuroticism, while contributing to personal and professional success, are seen as innate traits that are likely to remain static over time. Distinguishing EI from personality traits helps us better target the skills that can improve desirable outcomes (Brackett et al., 2013). Approaching EI with language that provides the opportunity for personal growth is crucial to its application. Because the ability model aligns with this approach, the remainder of this module will focus on ability EI and the ways in which it can be applied both in professional and academic settings.

Outcomes

Historically, emotions have been thought to have no place in the classroom or workplace (Sutton & Wheatly, 2003). Yet today, we know empirical research supports the belief that EI has the potential to influence decision making, health, relationships, and performance in both professional and academic settings (e.g., Brackett et al., 2013; Brackett, Rivers, & Salovey, 2011).

Workplace

Research conducted in the workplace supports positive links between EI and enhanced job

performance, occupational well-being, and leadership effectiveness. In one study, EI was associated with performance indicators such as company rank, percent merit increase, ratings of interpersonal facilitation, and affect and attitudes at work (Lopes, Grewal, Kadis, Gall, & Salovey, 2006). Similar correlations have been found between EI and a variety of managerial simulations involving problem solving, determining employee layoffs, adjusting claims, and negotiating successfully (Day & Carroll, 2004; Feyerherm & Rice, 2002; Mueller & Curhan, 2006). Emotion management is seen as most likely to affect job performance by influencing social and business interactions across a diverse range of industries (O'Boyle et al., 2010).



Think of a time when you made a choice to demonstrate emotional intelligence at work – what happened? [Image: Pricenfees, <https://goo.gl/8GOJPR>, CC BY 2.0, <https://goo.gl/BRvSA7>]

Leaders in the workplace also benefit from high EI. Experts in the field of organizational behavior are beginning to view leadership as a process of social interactions where leaders motivate, influence, guide, and empower followers to achieve organizational goals (Bass & Riggio, 2006). This is known as transformational leadership—where leaders create a vision and then inspire others to work in this direction (Bass, 1985). In a sample of 24 managers, MSCEIT scores correlated positively with a leader's ability to inspire followers to emulate their own actions and attend to the needs and problems of each individual (Leban & Zulauf, 2004).

Schools

When applied in educational settings, theoretical foundations of EI are often integrated into **social and emotional learning (SEL)** programs. SEL is the process of merging thinking, feeling, and behaving. These skills enable individuals to be aware of themselves and of others, make responsible decisions, and manage their own behaviors and those of others (Elias et al., 1997; Elbertson, Brackett, & Weissberg, 2010). SEL programs are designed to enhance the climate of a classroom, school, or district, with the ultimate goal of enhancing children's social and emotional skills and improving their academic outcomes (Greenberg et al., 2003). Adopting curricula that focus on these elements is believed to enable success in academics,

relationships, and, ultimately, in life (Becker & Luthar, 2002; Catalino, Berglundh, Ryan, Lonczek, & Hawkins, 2004).

Take a moment to think about the role of a teacher. How might emotions impact the climate of a classroom? If a teacher enters a classroom feeling anxious, disgruntled, or unenthused, these states will most likely be noticed, and felt, by the students. If not managed well, these negative emotions can hurt the classroom dynamic and prevent student learning (Travers, 2001). Research suggests that the abilities to perceive, use, understand, and manage emotions are imperative for effective teaching (Reyes, Brackett, Rivers, White, & Salovey, 2012; Brackett, Reyes, Rivers, Elbertson, & Salovey, 2011; Hargreaves, 2001). In a study that examined the relationship between emotion regulation and both job satisfaction and burnout among secondary-school teachers, researchers found that emotion regulation among teachers was associated with positive affect, support from principals, job satisfaction, and feelings of personal accomplishment (Brackett, Palomera, Mojsa-Kaja, Reyes, & Salovey, 2010).

EI, when embedded into SEL programs, has been shown to contribute positively to personal and academic success in students (Durlak, Weissberg, Dymnicki, Tayloer, & Schellinger, 2011). Research also shows that strong emotion regulation can help students pay attention in class, adjust to the school environment, and manage academic anxiety (Lopes & Salovey, 2004; Mestre, Guil, Lopes, Salovey, & Gil-Olarte, 2006). A recent randomized control trial of RULER* also found that, after one year, schools that used RULER—compared with those that used only the standard curriculum—were rated by independent observers as having higher degrees of warmth and connectedness between teachers and students, more autonomy and leadership, less bullying among students, and teachers who focused more on students' interests and motivations (Rivers, Brackett, Reyes, Elbertson, & Salovey, 2013).

*RULER - Recognize emotions in oneself and in other people. Understand the causes and consequences of a wide range of emotions. Label emotions using a sophisticated vocabulary. Express emotions in socially appropriate way. Regulate emotions effectively.

Limitations and Future Directions

There is a need for further development in EI theory and measurement, as well as more empirical research on its associated outcomes (Mayer, Salovey, & Caruso, 2008). Despite its prominent role as the signature performance assessment of EI, the MSCEIT has a number of limitations. For example, it does not allow for the assessment of several abilities. These abilities include the expression of emotion and monitoring or reflecting on one's own emotions. (Brackett et al. 2013). Researchers must also address growing criticisms, particularly



Future directions for EI research include more study of those in cultures outside North America and Europe, and more attention to the dynamics of EI in the workplace and schools. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

those that stretch beyond the measurement debate and question the validity of the EI construct when defined too broadly (Locke, 2005). In order to advance EI research, there is a great need for investigators to address these issues by reconciling disparate definitions and refining existing measures. Potential considerations for future research in the field should include deeper investigation into the genetic (versus acquired) and fluid (versus crystallized) aspects of EI. The cultural implications and differences of EI also are important to consider. Studies should expand beyond the United States and Europe in order for the theory of EI to be cross-culturally valid and for its applications and outcomes to be achieved more universally. Greater attention should

also be paid to developmental trajectories, gender differences, and how EI operates in the workplace and educational settings (Brackett et al., 2013).

Although further explorations and research in the field of EI are needed, current findings indicate a fundamental relationship between emotion and cognition. Returning to our opening question, what will you do when denied concert tickets? One of the more compelling aspects of EI is that it grants us reign over our own emotions—forces once thought to rule the self by denying individual agency. But with this power comes responsibility. If you are enraged about not getting tickets to the show, perhaps you can take a few deep breaths, go for a walk, and wait until your physiological indicators (shaky hands or accelerated heartbeat) subside. Once you've removed yourself, your feeling of rage may lessen to annoyance. Lowering the intensity level of this feeling (a process known as *down regulating*) will help re-direct your focus on the situation itself, rather than the activated emotion. In this sense, emotion regulation allows you to objectively view the point of conflict without dismissing your true feelings. Merely down regulating the emotional experience facilitates better problem solving. Now that you are less activated, what is the best approach? Should you talk to the ticket clerk? Ask to see the sales manager? Or do you let the group know how you felt when they cut the line? All of these options present better solutions than impulsively acting out rage.

As discussed in this module, research shows that the cultivation and development of EI

contributes to more productive, supportive, and healthy experiences. Whether we're waiting in a crowded public place, delivering lesson plans, or engaging in conversation with friends, we are the ultimate decision makers when it comes how we want to feel and, in turn, behave. By engaging the right mental processes and strategies, we can better understand, regulate, and manage our emotional states in order to live the lives we desire.

Outside Resources

Article: Are you emotionally intelligent? Here's how to know for sure. Inc.com Retrieved from:

<http://www.inc.com/travis-bradberry/are-you-emotionally-intelligent-here-s-how-to-know-for-sure.html>

Article: Grant, A. (2014, January 2). The dark side of emotional intelligence, The Atlantic. Retrieved from:

<http://www.theatlantic.com/health/archive/2014/01/the-dark-side-of-emotional-intelligence/282720/>

Article: Gregoire, C. (2014, January 23) How emotionally intelligent are you? Here's how to tell. Huffington Post. Retrieved from:

http://www.huffingtonpost.com/2013/12/05/are-you-emotionally-intel_n_4371920.html

Book: Goleman, D. (1995). Emotional intelligence. New York, NY: Bantam.

Book: Goleman, D. (1998). Working with emotional intelligence. New York, NY: Bantam.

Discussion Questions

1. What are the four emotional abilities that comprise EI, and how do they relate to each other?
2. What are three possible implications for using ability-based and mixed or trait-based models of EI?
3. Discuss the ways in which EI can contribute positively to the workplace and classroom settings.

Vocabulary

Ability model

An approach that views EI as a standard intelligence that utilizes a distinct set of mental abilities that (1) are intercorrelated, (2) relate to other extant intelligences, and (3) develop with age and experience (Mayer & Salovey, 1997).

Emotional intelligence

The ability to monitor one's own and others' feelings and emotions, to discriminate among them and to use this information to guide one's thinking and actions. (Salovey & Mayer, 1990). EI includes four specific abilities: perceiving, using, understanding, and managing emotions.

Four-Branch Model

An ability model developed by Drs. Peter Salovey and John Mayer that includes four main components of EI, arranged in hierarchical order, beginning with basic psychological processes and advancing to integrative psychological processes. The branches are (1) perception of emotion, (2) use of emotion to facilitate thinking, (3) understanding emotion, and (4) management of emotion.

Mayer-Salovey-Caruso Emotional Intelligence Test (MSCEIT)

A 141-item performance assessment of EI that measures the four emotion abilities (as defined by the four-branch model of EI) with a total of eight tasks.

Mixed and Trait Models

Approaches that view EI as a combination of self-perceived emotion skills, personality traits, and attitudes.

Performance assessment

A method of measurement associated with ability models of EI that evaluate the test taker's ability to solve emotion-related problems.

Self-report assessment

A method of measurement associated with mixed and trait models of EI, which evaluates the test taker's perceived emotion-related skills, distinct personality traits, and other characteristics.

Social and emotional learning (SEL)

The real-world application of EI in an educational setting and/or classroom that involves

curricula that teach the process of integrating thinking, feeling, and behaving in order to become aware of the self and of others, make responsible decisions, and manage one's own behaviors and those of others (Elias et al., 1997)

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Culture and Emotion

Jeanne Tsai

How do people's cultural ideas and practices shape their emotions (and other types of feelings)? In this module, we will discuss findings from studies comparing North American (United States, Canada) and East Asian (Chinese, Japanese, Korean) contexts. These studies reveal both cultural similarities and differences in various aspects of emotional life. Throughout, we will highlight the scientific and practical importance of these findings and conclude with recommendations for future research.

Learning Objectives

- Review the history of cross-cultural studies of emotion
- Learn about recent empirical findings and theories of culture and emotion
- Understand why cultural differences in emotion matter
- Explore current and future directions in culture and emotion research

Take a moment and imagine you are traveling in a country you've never been to before. Everything—the sights, the smells, the sounds—seems strange. People are speaking a language you don't understand and wearing clothes unlike yours. But they greet you with a smile and you sense that, despite the differences you observe, deep down inside these people have the same feelings as you. But is this true? Do people from opposite ends of the world really feel the same emotions? While most scholars agree that members of different cultures may vary in the foods they eat, the languages they speak, and the holidays they celebrate, there is disagreement about the extent to which culture shapes people's emotions and feelings

—including what people feel, what they express, and what they do during an emotional event. Understanding how culture shapes people's emotional lives and what impact emotion has on psychological health and well-being in different cultures will not only advance the study of human behavior but will also benefit multicultural societies. Across a variety of settings—academic, business, medical—people worldwide are coming into more contact with people from foreign cultures. In order to communicate and function effectively in such situations, we must understand the ways cultural ideas and practices shape our emotions.

Historical Background

In the 1950s and 1960s, social scientists tended to fall into either one of two camps. The universalist camp claimed that, despite cultural differences in customs and traditions, at a fundamental level all humans feel similarly. These universalists believed that emotions evolved as a response to the environments of our primordial ancestors, so they are the same across all cultures. Indeed, people often describe their emotions as “automatic,” “natural,” “physiological,” and “instinctual,” supporting the view that emotions are hard-wired and universal.



Universalists point to our prehistoric ancestors as the source of emotions that all humans share. [Image: Stefan Sheer, <https://goo.gl/x56mw9>, CC BY-SA 3.0, <https://goo.gl/tCiqIm>]

The social constructivist camp, however, claimed that despite a common evolutionary heritage, different groups of humans evolved to adapt to their distinctive environments. And because human environments vary so widely, people's emotions are also culturally variable. For instance, Lutz (1988) argued that many Western views of emotion assume that emotions are “singular events situated within individuals.” However, people from Ifaluk (a small island near Micronesia) view emotions as “exchanges between individuals” (p. 212). Social constructivists contended that because cultural ideas and practices are all-encompassing, people are often

unaware of how their feelings are shaped by their culture. Therefore emotions can feel automatic, natural, physiological, and instinctual, and yet still be primarily culturally shaped.

In the 1970s, Paul Ekman conducted one of the first scientific studies to address the universalist–social constructivist debate. He and Wallace Friesen devised a system to measure people’s facial muscle activity, called the Facial Action Coding System (FACS; Ekman & Friesen, 1978). Using FACS, Ekman and Friesen analyzed people’s facial expressions and identified specific facial muscle configurations associated with specific emotions, such as happiness, anger, sadness, fear, disgust. Ekman and Friesen then took photos of people posing with these different expressions (Figure 1). With the help of colleagues at different universities around the world, Ekman and Friesen showed these pictures to members of vastly different cultures, gave them a list of emotion words (translated into the relevant languages), and asked them to match the facial expressions in the photos with their corresponding emotion words on the list (Ekman & Friesen, 1971; Ekman et al., 1987).

Across cultures, participants “recognized” the emotional facial expressions, matching each picture with its “correct” emotion word at levels greater than chance. This led Ekman and his colleagues to conclude that there are universally recognized emotional facial expressions. At the same time, though, they found considerable variability across cultures in recognition rates. For instance, whereas 95% of U.S. participants associated a smile with “happiness,” only 69% of Sumatran participants did. Similarly, 86% of U.S. participants associated wrinkling of the nose with “disgust,” but only 60% of Japanese did (Ekman et al., 1987). Ekman and colleagues interpreted this variation as demonstrating cultural differences in “display rules,” or rules about what emotions are appropriate to show in a given situation (Ekman, 1972). Indeed, since this initial work, Matsumoto and his colleagues have demonstrated widespread cultural differences in display rules (Safdar et al., 2009). One prominent example of such differences is biting one’s tongue. In India, this signals embarrassment; however, in the U.S. this expression has no such meaning (Haidt & Keltner, 1999).

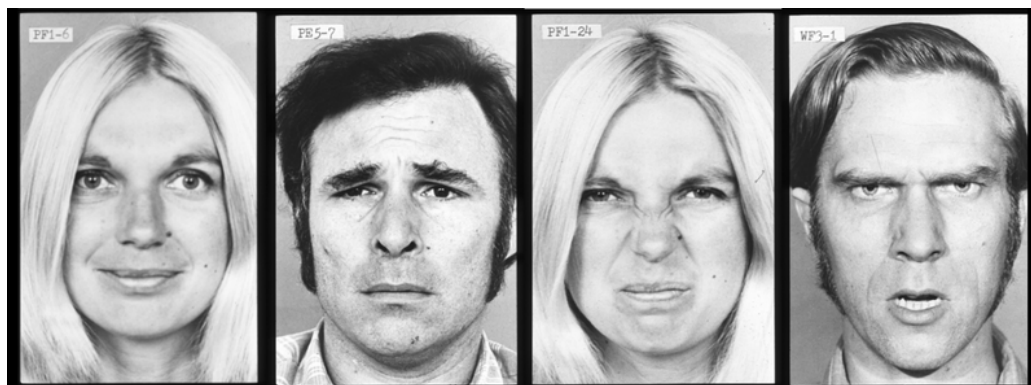


Photo credit - Paul Ekman Ph.D. / Paul Ekman Group, LLC.

Figure 1. Facial expressions associated with happiness, sadness, disgust, and anger based on the Facial Action Coding System. [Image: Paul Ekman, used with permission]

These findings suggest both cultural similarities and differences in the recognition of emotional facial expressions (although see Russell, 1994, for criticism of this work). Interestingly, since the mid-2000s, increasing research has demonstrated cultural differences not only in display rules, but also the degree to which people focus on the face (versus other aspects of the social context; Masuda, Ellsworth, Mesquita, Leu, Tanida, & Van de Veerdonk, 2008), and on different features of the face (Yuki, Maddux, & Matsuda, 2007) when perceiving others' emotions. For example, people from the United States tend to focus on the mouth when interpreting others' emotions, whereas people from Japan tend to focus on the eyes.

But how does culture shape other aspects of emotional life—such as how people emotionally respond to different situations, how they want to feel generally, and what makes them happy? Today, most scholars agree that emotions and other related states are multifaceted, and that cultural similarities and differences exist for each facet. Thus, rather than classifying emotions as *either* universal *or* socially-constructed, scholars are now attempting to identify the specific similarities and differences of emotional life across cultures. These endeavors are yielding new insights into the effects of cultural on emotion.

Current and Research Theory

Given the wide range of cultures and facets of emotion in the world, for the remainder of the module we will limit our scope to the two cultural contexts that have received the most empirical attention by social scientists: North America (United States, Canada) and East Asia (China, Japan, and Korea). Social scientists have focused on North American and East Asian contexts because they differ in obvious ways, including their geographical locations, histories, languages, and religions. Moreover, since the 1980s large-scale studies have revealed that North American and East Asian contexts differ in their overall values and attitudes, such as the prioritization of personal vs. group needs (individualism vs. collectivism; Hofstede, 2001). Whereas North American contexts encourage members to prioritize personal over group needs (to be “individualistic”), East Asian contexts encourage members to prioritize group over personal needs (to be “collectivistic”).

Cultural Models of Self in North American and East Asian Contexts

In a landmark paper, cultural psychologists Markus and Kitayama (1991) proposed that previously observed differences in individualism and collectivism translated into different models of the self—or one's personal concept of who s/he is as a person. Specifically, the researchers argued that in North American contexts, the dominant model of the self is an independent one, in which being a person means being distinct from others and behaving

accordingly across situations. In East Asian contexts, however, the dominant model of the self is an interdependent one, in which being a person means being fundamentally connected to others and being responsive to situational demands. For example, in a classic study (Cousins, 1989), American and Japanese students were administered the Twenty Statements Test, in which they were asked to complete the sentence stem, "I am _____," twenty times. U.S. participants were more likely than Japanese participants to complete the stem with psychological attributes (e.g., friendly, cheerful); Japanese participants, on the other hand, were more likely to complete the stem with references to social roles and responsibilities (e.g., a daughter, a student) (Cousins, 1989). These different models of the self result in different principles for interacting with others. An independent model of self teaches people to express themselves and try to influence others (i.e., change their environments to be consistent with their own beliefs and desires). In contrast, an interdependent model of self teaches people to suppress their own beliefs and desires and adjust to others' (i.e., fit in with their environment) (Heine, Lehman, Markus, & Kitayama, 1999; Morling, Kitayama, & Miyamoto, 2002; Weisz, Rothbaum, & Blackburn, 1984). Markus and Kitayama (1991) argue that these different models of self have significant implications for how people in Western and East Asian contexts feel.

Cultural Similarities and Differences in Emotion: Comparisons of North American and East Asian Contexts

A considerable body of empirical research suggests that these different models of self shape various aspects of emotional dynamics. Next we will discuss several ways culture shapes emotion, starting with emotional response.

People's Physiological Responses to Emotional Events Are Similar Across Cultures, but Culture Influences People's Facial Expressive Behavior

How does culture influence people's responses to emotional events? Studies of emotional response tend to focus on three components: physiology (e.g., how fast one's heart beats), subjective experience (e.g., feeling intensely happy or sad), and facial expressive behavior (e.g., smiling or frowning). Although only a few studies have simultaneously measured these different aspects of emotional response, those that do tend to observe more similarities than differences in physiological responses between cultures. That is, regardless of culture, people tend to respond similarly in terms of physiological (or bodily) expression. For instance, in one study, European American and Hmong (pronounced "muhng") American participants were

asked to relive various emotional episodes in their lives (e.g., when they lost something or someone they loved; when something good happened) (Tsai, Chentsova-Dutton, Freire-Bebeau, & Prymus, 2002). At the level of physiological arousal (e.g., heart rate), there were no differences in how the participants responded. However, their facial expressive behavior told a different story. When reliving events that elicited happiness, pride, and love, European Americans smiled more frequently and more intensely than did their Hmong counterparts—though all participants reported feeling happy, proud, and in love at similar levels of intensity. And similar patterns have emerged in studies comparing European Americans with Chinese Americans during different emotion-eliciting tasks (Tsai et al., 2002; Tsai, Levenson, & McCoy, 2006; Tsai, Levenson, & Carstensen, 2000). Thus, while the physiological aspects of emotional responses appear to be similar across cultures, their accompanying facial expressions are more culturally distinctive.



Although study participants from different cultural backgrounds reported similar emotions and levels of intensity when recalling important episodes in their lives, there were significant differences in facial expressions in response to those emotions. [Image: Andrew Sweeney, <https://goo.gl/Npc7Wm>, CC BY-NC-SA 4.0, <https://goo.gl/H2QaA8>]

Again, these differences in facial expressions during positive emotional events are consistent with findings from cross-cultural studies of *display rules*, and stem from the models of self-description discussed above: In North American contexts that promote an **independent self**, individuals tend to express their emotions to influence others. Conversely, in East Asian contexts that promote an interdependent self, individuals tend to control and suppress their emotions to adjust to others.

People Suppress Their Emotions Across Cultures, but Culture Influences the Consequences of Suppression for Psychological Well-Being

If the cultural ideal in North American contexts is to express oneself, then suppressing emotions (not showing how one feels) should have negative consequences. This is the

assumption underlying hydraulic models of emotion: the idea that emotional suppression and repression impair psychological functioning (Freud, 1910). Indeed, significant empirical research shows that suppressing emotions can have negative consequences for psychological well-being in North American contexts (Gross, 1998). However, Soto and colleagues (2011) find that the relationship between suppression and psychological well-being varies by culture. True, with European Americans, emotional suppression is associated with higher levels of depression and lower levels of life satisfaction. (Remember, in these individualistic societies, the expression of emotion is a fundamental aspect of positive interactions with others.) On the other hand, since for Hong Kong Chinese, emotional suppression is needed to adjust to others (in this interdependent community, suppressing emotions is how to appropriately interact with others), it is simply a part of normal life and therefore not associated with depression or life satisfaction.

These findings are consistent with research suggesting that factors related to clinical depression vary between European Americans and Asian Americans. European Americans diagnosed with depression show dampened or muted emotional responses (Bylsma, Morris, & Rottenberg, 2008). For instance, when shown sad or amusing film clips, depressed European Americans respond less intensely than their nondepressed counterparts. However, other



Someone from a collectivist culture is more likely to think about how their own accomplishments might impact others. An otherwise positive achievement for one person could cause another to feel something negative, with mixed emotions as the result. [Image: lian xiaoxiao, <https://goo.gl/js5jDw>, CC BY-SA 2.0, <https://goo.gl/jSSrcO>]

studies have shown that depressed East Asian Americans (i.e., people of East Asian descent who live in the United States) demonstrate *similar or increased* emotional responses compared with their nondepressed counterparts (Chentsova-Dutton et al., 2007; Chentsova-Dutton, Tsai, & Gotlib, 2010). In other words, depressed European Americans show reduced emotional expressions, but depressed East Asian Americans do *not*—and, in fact, may express *more* emotion. Thus, muted responses (which resemble suppression) are associated with depression in European American contexts, but not in East Asian contexts.

People Feel Good During Positive Events, but Culture Influences Whether People

Feel Bad During Positive Events

What about people's subjective emotional experiences? Do people across cultures *feel* the same emotions in similar situations, despite how they show them? Recent studies indicate that culture affects whether people are likely to feel bad during good events. In North American contexts, people rarely feel bad after good experiences. However, a number of research teams have observed that, compared with people in North American contexts, people in East Asian contexts are more likely to feel bad *and* good ("mixed" emotions) during positive events (e.g., feeling worried after winning an important competition; Miyamoto, Uchida, & Ellsworth, 2010). This may be because, compared with North Americans, East Asians engage in more dialectical thinking (i.e., they are more tolerant of contradiction and change). Therefore, they accept that positive and negative feelings can occur simultaneously. In addition, whereas North Americans value maximizing positive states and minimizing negative ones, East Asians value a greater balance between the two (Sims, Tsai, Wang, Fung, & Zhang, 2013). To better understand this, think about how you would feel after getting the top score on a test that's graded on a curve. In North American contexts, such success is considered an individual achievement and worth celebrating. But what about the other students who will now receive a lower grade because you "raised the curve" with your good grade? In East Asian contexts, not only would students be more thoughtful of the overall group's success, but they would also be more comfortable acknowledging both the positive (their own success on the test) and the negative (their classmates' lower grades).

Again, these differences can be linked to cultural differences in models of the self. An interdependent model encourages people to think about how their accomplishments might affect others (e.g., make others feel bad or jealous). Thus, awareness of negative emotions during positive events may discourage people from expressing their excitement and standing out (as in East Asian contexts). Such emotional suppression helps individuals feel in sync with those around them. An independent model, however, encourages people to express themselves and stand out, so when something good happens, they have no reason to feel bad.

So far, we have reviewed research that demonstrates cultural similarities in physiological responses and in the ability to suppress emotions. We have also discussed the cultural differences in facial expressive behavior and the likelihood of experiencing negative feelings during positive events. Next, we will explore how culture shapes people's ideal or desired states.

People Want to Feel Good Across Cultures, but Culture Influences the Specific Good States People Want to Feel (Their "Ideal Affect")

Everyone welcomes positive feelings, but cultures vary in the specific types of **positive affective states** (see Figure 2) their people favor. An affective state is essentially the type of emotional arousal one feels coupled with its intensity—which can vary from pleasant to unpleasant (e.g., happy to sad), with high to low arousal (e.g., energetic to passive). Although people of all cultures experience this range of affective states, they can vary in their preferences for each. For example, people in North American contexts lean toward feeling excited, enthusiastic, energetic, and other “high arousal positive” states. People in East Asian contexts, however, generally prefer feeling calm, peaceful, and other “low arousal positive” states (Tsai, Knutson, & Fung, 2006). These cultural differences have been observed in young children between the ages of 3 and 5, college students, and adults between the ages of 60 and 80 (Tsai, Louie, Chen, & Uchida, 2007; Tsai, Sims, Thomas, & Fung, 2013), and are reflected in widely-distributed cultural products. For example, wherever you look in American contexts—women’s magazines, children’s storybooks, company websites, and even Facebook profiles (Figure 3)—you will find more open, excited smiles and fewer closed, calm smiles compared to Chinese contexts (Chim, Moon, Ang, Tsai, 2013; Tsai, 2007; Tsai, Louie, et al., 2007).

Two-Dimensional Map of Affective States

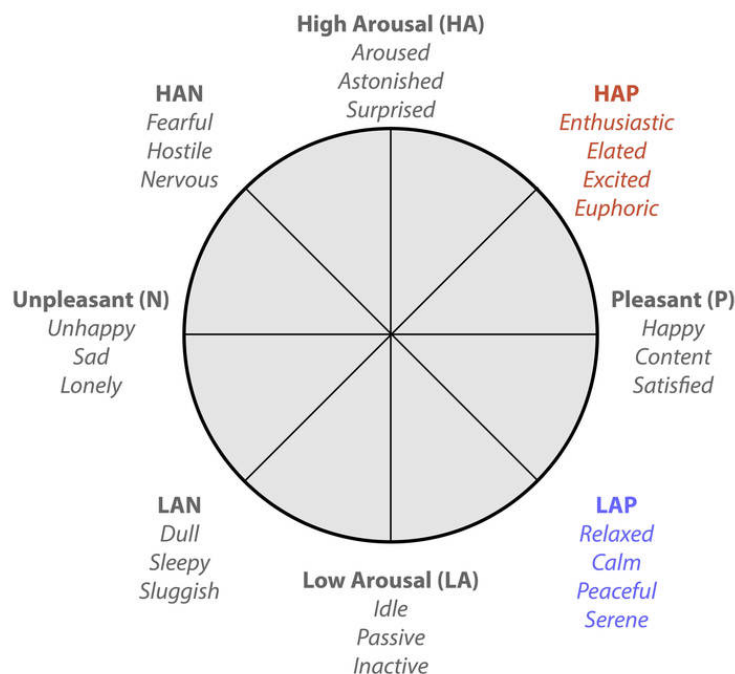


Figure 2: Adapted from Feldman, Barrett, and Russell (1999); Larsen and Diener ((1992); Russell (1991); Thayer (1989); Watson and Tellegen (1985)

Again, these differences in ideal affect (i.e., the emotional states that people believe are best) correspond to the independent and interdependent models described earlier: Independent selves want to influence others, which requires action (*doingsomething*), and action involves high arousal states. Conversely, interdependent selves want to adjust to others, which requires *suspending* action and attending to others—both of which involve low arousal states. Thus, the more that individuals and cultures want to influence others (as in North American contexts), the more they value excitement, enthusiasm, and other high arousal positive states. And, the more that individuals and cultures want to adjust to others (as in East Asian contexts), the more they value calm, peacefulness, and other low arousal positive states (Tsai, Miao, Seppala, Fung, & Yeung, 2007).



Figure 3: Sample Hong Kong Chinese (left) and European American (right) Facebook pages.

Because one's ideal affect functions as a guide for behavior and a way of evaluating one's emotional states, cultural differences in ideal affect can result in different emotional lives. For example, several studies have shown that people engage in activities (e.g., recreational pastimes, musical styles) consistent with their cultural ideal affect. That is, people from North American contexts (who value high arousal affective states) tend to prefer thrilling activities like skydiving, whereas people from East Asian contexts (who value low arousal affective states) prefer tranquil activities like lounging on the beach (Tsai, 2007). In addition, people base their conceptions of well-being and happiness on their ideal affect. Therefore, European Americans are more likely to define well-being in terms of excitement, whereas Hong Kong Chinese are more likely to define well-being in terms of calmness. Indeed, among European Americans, the less people experience *high* arousal positive states, the more depressed they are. But, among Hong Kong Chinese—you guessed it!—the less people experience *low* arousal positive states, the more depressed they are (Tsai, Knutson, & Fung, 2006).

People Base Their Happiness on Similar Factors Across Cultures, but Culture Influences the Weight Placed on Each Factor



Research has shown that self-esteem is more highly correlated with life satisfaction in individualistic cultures than in collectivist cultures. [Image: Erik, <https://goo.gl/N8zccv>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

What factors make people happy or satisfied with their lives? We have seen that discrepancies between how people actually feel (actual affect) and how they want to feel (ideal affect)—as well as people's suppression of their ideal affect—are associated with depression. But happiness is based on other factors as well. For instance, Kwan, Bond, & Singelis (1997) found that while European Americans and Hong Kong Chinese subjects both based life satisfaction on how they felt about themselves (self-esteem) and their relationships (relationship harmony), their weighting of each factor was different. That is, European Americans based their life satisfaction primarily on self-esteem, whereas Hong Kong Chinese based their life satisfaction equally on self-esteem *and* relationship harmony. Consistent with these findings,

Oishi and colleagues (1999) found in a study of 39 nations that self-esteem was more strongly correlated with life satisfaction in more individualistic nations compared to more collectivistic ones. Researchers also found that in individualistic cultures people rated life satisfaction based on their emotions more so than on social definitions (or norms). In other words, rather than using social norms as a guideline for what constitutes an ideal life, people in individualistic cultures tend to evaluate their satisfaction according to how they feel emotionally. In collectivistic cultures, however, people's life satisfaction tends to be based on a balance between their emotions and norms (Suh, Diener, Oishi, & Triandis, 1998). Similarly, other researchers have recently found that people in North American contexts are more likely to feel negative when they have poor mental and physical health, while people in Japanese contexts don't have this association (Curhan et al., 2013).

Again, these findings are consistent with cultural differences in models of the self. In North American, independent contexts, feelings about the self matter more, whereas in East Asian, interdependent contexts, feelings about others matter as much as or even more than feelings

about the self.

Why Do Cultural Similarities And Differences In Emotion Matter?

Understanding cultural similarities and differences in emotion is obviously critical to understanding emotions in general, and the flexibility of emotional processes more specifically. Given the central role that emotions play in our interaction, understanding cultural similarities and differences is especially critical to preventing potentially harmful miscommunications. Although misunderstandings are unintentional, they can result in negative consequences—as we’ve seen historically for ethnic minorities in many cultures. For instance, across a variety of North American settings, Asian Americans are often characterized as too “quiet” and “reserved,” and these low arousal states are often misinterpreted as expressions of disengagement or boredom—rather than expressions of the ideal of calmness. Consequently, Asian Americans may be perceived as “cold,” “stoic,” and “unfriendly,” fostering stereotypes of Asian Americans as “perpetual foreigners” (Cheryan & Monin, 2005). Indeed, this may be one reason Asian Americans are often overlooked for top leadership positions (Hyun, 2005).

In addition to averting cultural miscommunications, recognizing cultural similarities and differences in emotion may provide insights into other paths to psychological health and well-being. For instance, findings from a recent series of studies suggest that calm states are easier to elicit than excited states, suggesting that one way of increasing happiness in cultures that value excitement may be to increase the value placed on calm states (Chim, Tsai, Hogan, & Fung, 2013).

Current Directions In Culture And Emotion Research

What About Other Cultures?

In this brief review, we’ve focused primarily on comparisons between North American and East Asian contexts because most of the research in cultural psychology has focused on these comparisons. However, there are obviously a multitude of other cultural contexts in which emotional differences likely exist. For example, although Western contexts are similar in many ways, specific Western contexts (e.g., American vs. German) also differ from each other in substantive ways related to emotion (Koopmann-Holm & Matsumoto, 2011). Thus, future research examining other cultural contexts is needed. Such studies may also reveal additional, uninvestigated dimensions or models that have broad implications for emotion. In addition,

because more and more people are being raised within multiple cultural contexts (e.g., for many Chinese Americans, a Chinese immigrant culture at home and mainstream American culture at school), more research is needed to examine how people negotiate and integrate these different cultures in their emotional lives (for examples, see De Leersnyder, Mesquita, & Kim, 2011; Perunovic, Heller, & Rafaeli, 2007).

How Are Cultural Differences in Beliefs About Emotion Transmitted?

According to Kroeber and Kluckhohn (1952), cultural ideas are reflected in and reinforced by practices, institutions, and products. As an example of this phenomenon—and illustrating the point regarding cultural differences in ideal affect—bestselling children’s storybooks in the United States often contain more exciting and less calm content (smiles and activities) than do bestselling children’s storybooks in Taiwan (Tsai, Louie, et al., 2007). To investigate this further, the researchers randomly assigned European American, Asian American, and Taiwanese Chinese preschoolers to be read either stories with exciting content or stories with calm content. Across all of these cultures, the kids who were read stories with exciting content were afterward more likely to value excited states, whereas those who were read stories with calm content were more likely to value calm states. As a test, after hearing the stories, the kids were shown a list of toys and asked to select their favorites. Those who heard the exciting stories wanted to play with more arousing toys (like a drum that beats loud and fast), whereas those who heard the calm stories wanted to play with less arousing toys (like a drum that beats quiet and slow). These findings suggest that regardless of ethnic background, direct exposure to storybook content alters children’s ideal affect. More studies are needed to assess whether a similar process occurs when children and adults are chronically exposed to various types of cultural products. As well, future studies should examine other ways cultural ideas regarding emotion are transmitted (e.g., via interactions with parents and teachers).



Children's story books offer one interesting and effective way to study how early influences can impact a person's ideal affect. [Image: Vernon Barford School Library, <https://goo.gl/fghcae>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

Could These Cultural Differences Be Due to Temperament?

An alternative explanation for cultural differences in emotion is that they are due to temperamental factors—that is, biological predispositions to respond in certain ways. (Might European Americans just be more emotional than East Asians because of genetics?) Indeed, most models of emotion acknowledge that both culture *and* temperament play roles in emotional life, yet few if any models indicate how. Nevertheless, most researchers believe that despite genetic differences in founder populations (i.e., the migrants from a population who leave to create their own societies), culture has a greater impact on emotions. For instance, one theoretical framework, Affect Valuation Theory, proposes that cultural factors shape how people want to feel (“ideal affect”) more than how they actually feel (“actual affect”); conversely, temperamental factors influence how people actually feel more than how they want to feel (Tsai, 2007) (see Figure 4).

To test this hypothesis, European American, Asian American, and Hong Kong Chinese participants completed measures of temperament (i.e., stable dispositions, such as neuroticism or extraversion), actual affect (i.e., how people actually feel in given situations), ideal affect (i.e., how people would like to feel in given situations), and influential cultural values (i.e., personal beliefs transmitted through culture). When researchers analyzed the participants’ responses, they found that differences in ideal affect between cultures were associated more with cultural factors than with temperamental factors (Tsai, Knutson, & Fung, 2006). However, when researchers examined actual affect, they found this to be reversed: actual affect was more strongly associated with temperamental factors than cultural factors. Not all of the studies described above have ruled out a temperamental explanation, though, and more studies are needed to rule out the possibility that the observed group differences

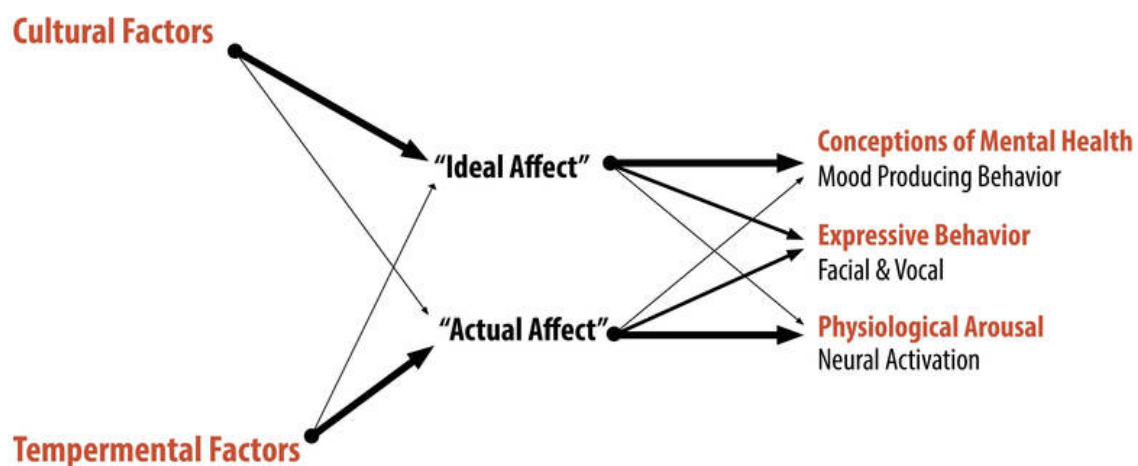


Figure 4: Affect valuation theory. Thicker lines indicate stronger predicted relationships.

are due to genetic factors instead of, or in addition to, cultural factors. Moreover, future studies should examine whether the links between temperament and emotions might vary across cultures, and how cultural and temperamental factors work together to shape emotion.

Summary

Based on studies comparing North American and East Asian contexts, there is clear evidence for cultural similarities and differences in emotions, and most of the differences can be traced to different cultural models of the self.

Consider your own concept of self for a moment. What kinds of pastimes do you prefer—activities that make you excited, or ones that make you calm? What kinds of feelings do you strive for? What is your ideal affect? Because emotions seem and feel so instinctual to us, it's hard to imagine that the way we experience them and the ones we desire are anything other than biologically programmed into us. However, as current research has shown (and as future research will continue to explore), there are myriad ways in which culture, both consciously and unconsciously, shapes people's emotional lives.

Outside Resources

Audio Interview: The Really Big Questions “What Are Emotions?” Interview with Paul Ekman, Martha Nussbaum, Dominique Moisi, and William Reddy

http://www.trbq.org/index.php?option=com_content&view=category&layout=blog&id=16&Itemid=43

Book: Ed Diener and Robert Biswas-Diener: Happiness: Unlocking the Mysteries of Psychological Wealth

Book: Eric Weiner: The Geography of Bliss

Book: Eva Hoffmann: Lost in Translation: Life in a New Language

Book: Hazel Markus: Clash: 8 Cultural Conflicts That Make Us Who We Are

Video: Social Psychology Alive

<http://psychology.stanford.edu/~tsailab/PDF/socpsychalive.wmv>

Video: The Really Big Questions “Culture and Emotion,” Dr. Jeanne Tsai

<http://youtu.be/RQaEaUwNoiw>

Video: Tsai’s description of cultural differences in emotion

<http://youtu.be/T46EZ8LH8Ss>

Web: Acculturation and Culture Collaborative at Leuven

<http://ppw.kuleuven.be/home/english/research/cscp/acc-research>

Web: Culture and Cognition at the University of Michigan

<http://culturecognition.isr.umich.edu/>

Web: Experts In Emotion Series, Dr. June Gruber, Department of Psychology, Yale University

http://www.yalepeplab.com/teaching/psych131_summer2013/expertseries.php

Web: Georgetown Culture and Emotion Lab

<http://georgetownculturelab.wordpress.com/>

Web: Paul Ekman’s website

<http://www.paulekman.com>

Web: Penn State Culture, Health, and Emotion Lab

<http://www.personal.psu.edu/users/m/r/mrm280/sotosite/>

Web: Stanford Culture and Emotion Lab

<http://www-psych.stanford.edu/~tsailab/index.htm>

Web: Wesleyan Culture and Emotion Lab

<http://culture-and-emotion.research.wesleyan.edu/>

Discussion Questions

1. What cultural ideas and practices related to emotion were you exposed to when you were a child? What cultural ideas and practices related to emotion are you currently exposed to as an adult? How do you think they shape your emotional experiences and expressions?
2. How can researchers avoid inserting their own beliefs about emotion in their research?
3. Most of the studies described above are based on self-report measures. What are some of the advantages and disadvantages of using self-report measures to understand the cultural shaping of emotion? How might the use of other behavioral methods (e.g., neuroimaging) address some of these limitations?
4. Do the empirical findings described above change your beliefs about emotion? How?
5. Imagine you are a manager of a large American company that is beginning to do work in China and Japan. How will you apply your current knowledge about culture and emotion to prevent misunderstandings between you and your Chinese and Japanese employees?

Vocabulary

Affect

Feelings that can be described in terms of two dimensions, the dimensions of arousal and valence (Figure 2). For example, high arousal positive states refer to excitement, elation, and enthusiasm. Low arousal positive states refer to calm, peacefulness, and relaxation. Whereas “actual affect” refers to the states that people actually feel, “ideal affect” refers to the states that people ideally want to feel.

Culture

Shared, socially transmitted ideas (e.g., values, beliefs, attitudes) that are reflected in and reinforced by institutions, products, and rituals.

Emotions

Changes in subjective experience, physiological responding, and behavior in response to a meaningful event. Emotions tend to occur on the order of seconds (in contrast to moods which may last for days).

Feelings

A general term used to describe a wide range of states that include emotions, moods, traits and that typically involve changes in subjective experience, physiological responding, and behavior in response to a meaningful event. Emotions typically occur on the order of seconds, whereas moods may last for days, and traits are tendencies to respond a certain way across various situations.

Independent self

A model or view of the self as distinct from others and as stable across different situations. The goal of the independent self is to express and assert the self, and to influence others. This model of self is prevalent in many individualistic, Western contexts (e.g., the United States, Australia, Western Europe).

Interdependent self

A model or view of the self as connected to others and as changing in response to different situations. The goal of the interdependent self is to suppress personal preferences and desires, and to adjust to others. This model of self is prevalent in many collectivistic, East Asian contexts (e.g., China, Japan, Korea).

Social constructivism

Social constructivism proposes that knowledge is first created and learned within a social context and is then adopted by individuals.

Universalism

Universalism proposes that there are single objective standards, independent of culture, in basic domains such as learning, reasoning, and emotion that are a part of all human experience.

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Knowledge Emotions: Feelings that Foster Learning, Exploring, and Reflecting

Paul Silvia

When people think of emotions they usually think of the obvious ones, such as happiness, fear, anger, and sadness. This module looks at the knowledge emotions, a family of emotional states that foster learning, exploring, and reflecting. Surprise, interest, confusion, and awe come from events that are unexpected, complicated, and mentally challenging, and they motivate learning in its broadest sense, be it learning over the course of seconds (finding the source of a loud crash, as in surprise) or over a lifetime (engaging with hobbies, pastimes, and intellectual pursuits, as in interest). The module reviews research on each emotion, with an emphasis on causes, consequences, and individual differences. As a group, the knowledge emotions motivate people to engage with new and puzzling things rather than avoid them. Over time, engaging with new things, ideas, and people broadens someone's experiences and cultivates expertise. The knowledge emotions thus don't gear up the body like fear, anger, and happiness do, but they do gear up the mind—a critical task for humans, who must learn essentially everything that they know.

Learning Objectives

- Identify the four knowledge emotions.
- Describe the patterns of appraisals that bring about these emotions.
- Discuss how the knowledge emotions promote learning.
- Apply the knowledge emotions to enhancing learning and education, and to one's own life.

Introduction

What comes to mind when you think of emotions? It's probably the elation of happiness, the despair of sadness, or the freak-out fright of fear. Emotions such as happiness, anger, sadness, and fear are important emotions, but human emotional experience is vast—people are capable of experiencing a wide range of feelings.

This module considers the **knowledge emotions**, a profoundly important family of emotions associated with learning, exploring, and reflecting. The family of knowledge emotions has four main members: *surprise*, *interest*, *confusion*, and *awe*. These are considered knowledge emotions for two reasons. First, the events that bring them about involve knowledge: These emotions happen when something violates what people expected or believed. Second, these emotions are fundamental to learning: Over time, they build useful knowledge about the world.



The knowledge emotions—surprise, interest, confusion, and awe—are what help to promote learning. For example, if something unusual happens (surprise), you study it to figure out why it happened. [Image: typexnick, <https://goo.gl/WIjrRR>, CC BY 2.0, <https://goo.gl/BRvSA7>]

Some Background About Emotions

Before jumping into the knowledge emotions, we should consider what emotions do and when emotions happen. According to **functionalist theories of emotion**, emotions help people manage important tasks (Keltner & Gross, 1999; Parrott, 2001). Fear, for example, mobilizes the body to fight or flee; happiness rewards achieving goals and builds attachments to other people. What do knowledge emotions do? As we'll see in detail later, they motivate *learning*, viewed in its broadest sense, during times that the environment is puzzling or erratic. Sometimes the learning is on a short time scale. Surprise, for example, makes people stop what they are doing, pay attention to the surprising thing, and evaluate whether it is dangerous (Simons, 1996). After a couple seconds, people have learned what they needed to know and get back to what they were doing. But sometimes the learning takes place over the lifespan. Interest, for example, motivates people to learn about things over days, weeks, and years.

Finding something interesting motivates “for its own sake” learning and is probably the major engine of human competence (Izard, 1977; Silvia, 2006).

What causes emotions to happen in the first place? Although it usually feels like something in the world—a good hug, a snake slithering across the driveway, a hot-air balloon shaped like a question mark—causes an emotion directly, emotion theories contend that emotions come from how we think about what is happening in the world, not what is literally happening. After all, if things in the world directly caused emotions, everyone would always have the same emotion in response to something. **Appraisal theories** (Ellsworth & Scherer, 2003; Lazarus, 1991) propose that each emotion is caused by a group of appraisals, which are evaluations and judgments of what events in the world mean for our goals and well-being: Is this relevant to me? Does it further or hinder my goals? Can I deal with it or do something about it? Did someone do it on purpose? Different emotions come from different answers to these appraisal questions.

With that as a background, in the following sections we’ll consider the nature, causes, and effects of each knowledge emotion. Afterward, we will consider some of their practical implications.

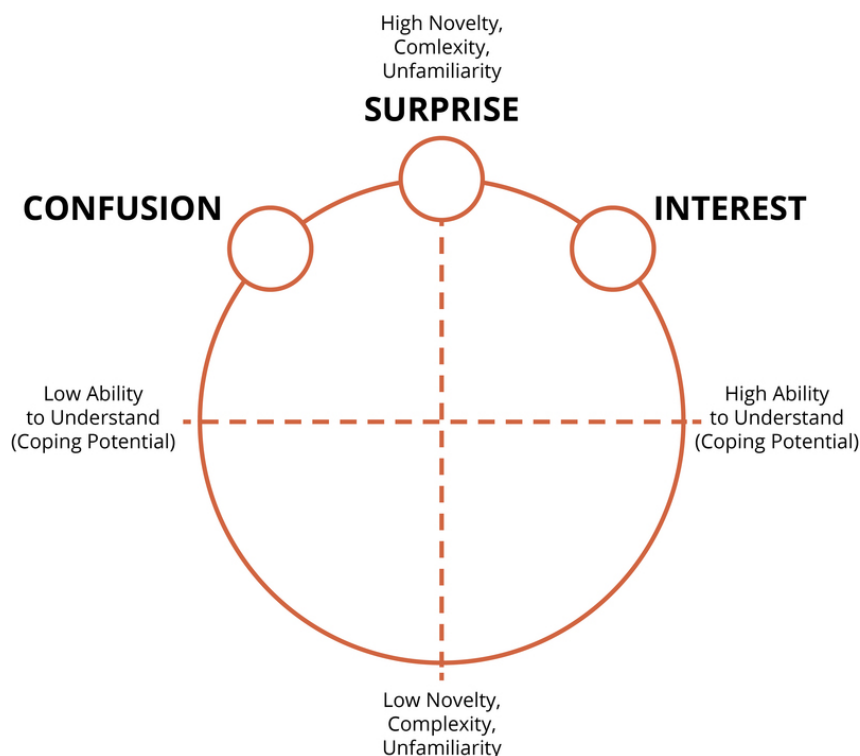


Figure 1. The appraisal space of surprise, interest, and confusion.

Surprise

Nothing gets people's attention like something startling. **Surprise**, a simple emotion, hijacks a person's mind and body and focuses them on a source of possible danger (Simons, 1996). When there's a loud, unexpected crash, people stop, freeze, and orient to the source of the noise. Their minds are wiped clean—after something startling, people usually can't remember what they had been talking about—and attention is focused on what just happened. By focusing all the body's resources on the unexpected event, surprise helps people respond quickly (Simons, 1996).

Surprise has only one appraisal: A single "expectedness check" (Scherer, 2001) seems to be involved. When an event is "high contrast"—it sticks out against the background of what people expected to perceive or experience—people become surprised (Berlyne, 1960; Teigen & Keren, 2003). Figure 1 shows this pattern visually: Surprise is high when unexpectedness is high.

Emotions are momentary states, but people vary in their propensity to experience them. Just as some people experience happiness, anger, and fear more readily, some people are much more easily surprised than others. At one end, some people are hard to surprise; at the other end, people are startled by minor noises, flashes, and changes. Like other individual differences in emotion, extreme levels of surprise propensity can be dysfunctional. When people have extreme surprise responses to mundane things—known as hyperstartling (Simons, 1996) and hyperekplexia (Bakker, van Dijk, van den Maagdenberg, & Tijssen, 2006)—everyday tasks such as driving or swimming become dangerous.

Interest

People are curious creatures. **Interest**—an emotion that motivates exploration and learning (Silvia, 2012)—is one of the most commonly experienced emotions in everyday life (Izard, 1977). Humans must learn virtually everything they know, from how to cook pasta to how the brain works, and interest is an engine of this massive undertaking of learning across the



Curiosity, a most familiar emotion, is the reason humans have developed and thrived as they have today. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

lifespan.

The function of interest is to engage people with things that are new, odd, or unfamiliar. Unfamiliar things can be scary or unsettling, which makes people avoid them. But if people always avoided new things they would learn and experience nothing. It's hard to imagine what life would be like if people weren't curious to try new things: We would never feel like watching a different movie, trying a different restaurant, or meeting new people. Interest is thus a counterweight to anxiety—by making unfamiliar things appealing, it motivates people to experience and think about new things. As a result, interest is an intrinsically motivated form of learning. When curious, people want to learn something for its own sake, to know it for the simple pleasure of knowing it, not for an external reward, such as learning to get money, impress a peer, or receive the approval of a teacher or parent.

Figure 1 shows the two appraisals that create interest. Like surprise, interest involves appraisals of novelty: Things that are unexpected, unfamiliar, novel, and complex can evoke interest (Berlyne, 1960; Hidi & Renninger, 2006; Silvia, 2008). But unlike surprise, interest involves an additional appraisal of coping potential. In appraisal theories, coping potential refers to people's evaluations of their ability to manage what is happening (Lazarus, 1991). When coping potential is high, people feel capable of handling the challenge at hand. For interest, this challenge is mental: Something odd and unexpected happened, and people can either feel able to understand it or not. When people encounter something that they appraise as both novel (high novelty and complexity) and comprehensible (high coping potential), they will find it interesting (Silvia, 2005).

The primary effect of interest is exploration: People will explore and think about the new and intriguing thing, be it an interesting object, person, or idea. By stimulating people to reflect and learn, interest builds knowledge and, in the long run, deep expertise. Consider, for example, the sometimes scary amount of knowledge people have about their hobbies. People who find cars, video games, high fashion, and soccer intrinsically interesting know an amazing amount about their passions—it would be hard to learn so much so quickly if people found it boring.

A huge amount of research shows that interest promotes learning that is faster, deeper, better, and more enjoyable (Hidi, 2001; Silvia, 2006). When people find material more interesting, they engage with it more deeply and learn it more thoroughly. This is true for simple kinds of learning—sentences and paragraphs are easier to remember when they are interesting (Sadoski, 2001; Schiefele, 1999)—and for broader academic success—people get better grades and feel more intellectually engaged in classes they find interesting (Krapp, 1999, 2002; Schiefele, Krapp, & Winteler, 1992).

Individual differences in interest are captured by trait curiosity (Kashdan, 2004; Kashdan et al., 2009). People low in curiosity prefer activities and ideas that are tried and true and familiar; people high in curiosity, in contrast, prefer things that are offbeat and new. Trait curiosity is a facet of openness to experience, a broader trait that is one of the five major factors of personality (McCrae, 1996; McCrae & Sutin, 2009). Not surprisingly, being high in openness to experience involves exploring new things and findings quirky things appealing. Research shows that curious, open people ask more questions in class, own and read more books, eat a wider range of food, and—not surprisingly, given their lifetime of engaging with new things—are a bit higher in intelligence (DeYoung, 2011; Kashdan & Silvia, 2009; Peters, 1978; Raine, Reynolds, Venables, & Mednick, 2002).

Confusion

Sometimes the world is weird. Interest is a wonderful resource when people encounter new and unfamiliar things, but those things aren't always comprehensible. Confusion happens when people are learning something that is both unfamiliar and hard to understand. In the appraisal space shown in Figure 1, confusion comes from appraising an event as high in novelty, complexity, and unfamiliarity as well as appraising it as hard to comprehend (Silvia, 2010, 2013).



Has anyone ever told you to “look up that answer for yourself?” The effort of working through your own confusion helps you learn the material better than simply being given the answer. [Image: CollegeDegrees360, <https://goo.gl/1Edneb>, CC BY-SA 2.0, <https://goo.gl/rxiUsF>]

Confusion, like interest, promotes thinking and learning. This isn't an obvious idea—our intuitions would suggest that confusion makes people frustrated and thus more likely to tune out and quit. But as odd as it sounds, making students confused can help them learn better. In an approach to learning known as impasse-driven learning (VanLehn, Siler, Murray, Yamauchi, & Baggett, 2003), making students confused motivates them to think through a problem instead of passively sitting and listening to what a teacher is saying. By actively thinking through the problem, students are learning actively and thus learning the material more deeply. In one experiment, for example, students learned about scientific research methods

from two virtual reality tutors (D’Mello, Lehman, Pekrun, & Graesser, in press). The tutors sometimes contradicted each other, however, which made the students confused. Measures of simple learning (memory for basic concepts) and deep learning (being able to transfer an idea to a new area) showed that students who had to work through confusion learned more deeply—they were better at correctly applying what they learned to new problems.

In a study of **facial expressions**, Rozin and Cohen (2003) demonstrated what all college teachers know: It’s easy to spot confusion on someone’s face. When people are confused, they usually furrow, scrunch, or lower their eyebrows and purse or bite their lips (Craig, D’Mello, Witherspoon, & Graesser, 2008; Durso, Geldbach, & Corballis, 2012). In a clever application of these findings, researchers have developed artificial intelligence (AI) teaching and tutoring systems that can detect expressions of confusion (Craig et al., 2008). When the AI system detects confusion, it can ask questions and give hints that help the student work through the problem.

Not much is known about individual differences related to confusion, but differences in how much people know are important. In one research study, people viewed short film clips from movies submitted to a local film festival (Silvia & Berg, 2011). Some of the people were film experts, such as professors and graduate students in media studies and film theory; others were novices, such as the rest of us who simply watch movies for fun. The experts found the clips much more interesting and much less confusing than the novices did. A similar study discovered that experts in the arts found experimental visual art more interesting and less confusing than novices did (Silvia, 2013).

Awe

Awe—a state of fascination and wonder—is the deepest and probably least common of the knowledge emotions. When people are asked to describe profound experiences, such as the experience of beauty or spiritual transformation, awe is usually mentioned (Cohen, Gruber, & Keltner, 2010). People are likely to report



Feeling awe often involves a sense of “feeling small,” like when you look up at the night sky or meet someone you really admire and respect. [Image: Kevin Dooley, <https://goo.gl/FEC1nE>, CC BY 2.0, <https://goo.gl/BRvSA7>]

experiencing awe when they are alone, engaged with art and music, or in nature (Shiota, Keltner, & Mossman, 2007).

Awe comes from two appraisals (Keltner & Haidt, 2003). First, people appraise something as vast, as beyond the normal scope of their experience. Thus, like the other knowledge emotions, awe involves appraising an event as inconsistent with one's existing knowledge, but the degree of inconsistency is huge, usually when people have never encountered something like the event before (Bonner & Friedman, 2011). Second, people engage in accommodation, which is changing their beliefs—about themselves, other people, or the world in general—to fit in the new experience. When something is massive (in size, scope, sound, creativity, or anything else) and when people change their beliefs to accommodate it, they'll experience awe.

A mild, everyday form of awe is chills, sometimes known as shivers or thrills. Chills involve getting goosebumps on the skin, especially the scalp, neck, back, and arms, usually as a wave that starts at the head and moves downward. Chills are part of strong awe experiences, but people often experience them in response to everyday events, such as compelling music and movies (Maruskin, Thrash, & Elliot, 2012; Nusbaum & Silvia, 2011). Music that evokes chills, for example, tends to be loud, have a wide frequency range (such as both low and high frequencies), and major dynamic shifts, such as a shift from quiet to loud or a shift from few to many instruments (Huron & Margulis, 2010).

Like the other knowledge emotions, awe motivates people to engage with something outside the ordinary. Awe is thus a powerful educational tool. In science education, it is common to motivate learning by inspiring wonder. One example comes from a line of research on astronomy education, which seeks to educate the public about astronomy by using awe-inspiring images of deep space (Arcand, Watzke, Smith, & Smith, 2010). When people see beautiful and striking color images of supernovas, black holes, and planetary nebulas, they usually report feelings of awe and wonder. These feelings then motivate them to learn about what they are seeing and their scientific importance (Smith et al., 2011).

Regarding individual differences, some people experience awe much more often than others. One study that developed a brief scale to measure awe—the items included statements such as “I often feel awe” and “I feel wonder almost every day”—found that people who often experience awe are much higher in openness to experience (a trait associated with openness to new things and a wide emotional range) and in extraversion (a trait associated with positive emotionality) (Shiota, Keltner, & John, 2006). Similar findings appear for when people are asked how often they experience awe in response to the arts (Nusbaum & Silvia, in press). For example, people who say that they often “feel a sense of awe and wonder” when listening to music are much higher in openness to experience (Silvia & Nusbaum, 2011).

Implications of the Knowledge Emotions



knowledge emotions help us form relationships and solve important problems. [Image: CERDEC, <https://goo.gl/KQL8M8>, CC BY 2.0, <https://goo.gl/BRvSA7>]

exploring, and reflecting.

Learning about the knowledge emotions expands our ideas about what emotions are and what they do. Emotions clearly play important roles in everyday challenges such as responding to threats and building relationships. But emotions also aid in other, more intellectual challenges for humans. Compared with other animals, we are born with little knowledge but have the potential for enormous intelligence. Emotions such as surprise, interest, confusion, and awe first signal that something awry has happened that deserves our attention. They then motivate us to engage with the new things that strain our understanding of the world and how it works. Emotions surely aid fighting and fleeing, but for most of the hours of most of our days, they mostly aid in learning,

Outside Resources

Video: A talk with Todd Kashdan, a well-known scholar in the field of curiosity and positive psychology, centered on curiosity

<http://www.youtube.com/watch?v=IPpVxZRqRc8>

Video: More from Todd

<http://www.youtube.com/watch?v=UunaTEpWrME>

Web: Aesthetics and Astronomy, a project that uses wonder and beauty to foster knowledge about the science of space

<http://astroart.cfa.harvard.edu/>

Web: The Emotion Computing Group, an interdisciplinary team that studies how to measure confusion and harness it for deeper learning, among other intriguing things

<https://sites.google.com/site/memphisemotivecomputing/Home>

Discussion Questions

1. Research shows that people learn more quickly and deeply when they are interested. Can you think of examples from your own life when you learned from interest versus from extrinsic rewards (e.g., good grades, approval from parents and peers)? Was learning more enjoyable or effective in one case?
2. How would you redesign a psychology lecture to harness the power of the knowledge emotions? How could you use interest, confusion, and awe to grab students' attention and motivate them to reflect and learn?
3. Psychology, like all the sciences, is fueled by wonder. For psychology, the wonder is about human nature and behavior. What, to you, is the most wondrous, amazing, and awe-inspiring idea or finding from the science of psychology? Does reflecting on this amazing fact motivate you to want to know more about it?
4. Many people only want to know something if it is practical—if it helps them get a job, make friends, find a mate, or earn money. But emotions such as interest and awe, by motivating learning for its own sake, often engage people in things that seem frivolous, silly, or impractical. What does this say about learning? Is some knowledge necessarily more valuable than other kinds?

Vocabulary

Accommodation

Changing one's beliefs about the world and how it works in light of new experience.

Appraisal structure

The set of appraisals that bring about an emotion.

Appraisal theories

Evaluations that relate what is happening in the environment to people's values, goals, and beliefs. Appraisal theories of emotion contend that emotions are caused by patterns of appraisals, such as whether an event furthers or hinders a goal and whether an event can be coped with.

Awe

An emotion associated with profound, moving experiences. Awe comes about when people encounter an event that is vast (far from normal experience) but that can be accommodated in existing knowledge.

Chills

A feeling of goosebumps, usually on the arms, scalp, and neck, that is often experienced during moments of awe.

Confusion

An emotion associated with conflicting and contrary information, such as when people appraise an event as unfamiliar and as hard to understand. Confusion motivates people to work through the perplexing information and thus fosters deeper learning.

Coping potential

People's beliefs about their ability to handle challenges.

Facial expressions

Part of the expressive component of emotions, facial expressions of emotion communicate inner feelings to others.

Functionalism theories of emotion

Theories of emotion that emphasize the adaptive role of an emotion in handling common problems throughout evolutionary history.

Impasse-driven learning

An approach to instruction that motivates active learning by having learners work through perplexing barriers.

Interest

An emotion associated with curiosity and intrigue, interest motivates engaging with new things and learning more about them. It is one of the earliest emotions to develop and a resource for intrinsically motivated learning across the life span.

Intrinsically motivated learning

Learning that is “for its own sake”—such as learning motivated by curiosity and wonder—instead of learning to gain rewards or social approval.

Knowledge emotions

A family of emotions associated with learning, reflecting, and exploring. These emotions come about when unexpected and unfamiliar events happen in the environment. Broadly speaking, they motivate people to explore unfamiliar things, which builds knowledge and expertise over the long run.

Openness to experience

One of the five major factors of personality, this trait is associated with higher curiosity, creativity, emotional breadth, and open-mindedness. People high in openness to experience are more likely to experience interest and awe.

Surprise

An emotion rooted in expectancy violation that orients people toward the unexpected event.

Trait curiosity

Stable individual-differences in how easily and how often people become curious.

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Motives and Goals

Ayelet Fishbach & Maferima Touré-Tillery

Your decisions and behaviors are often the result of a goal or motive you possess. This module provides an overview of the main theories and findings on goals and motivation. We address the origins, manifestations, and types of goals, and the various factors that influence motivation in goal pursuit. We further address goal conflict and, specifically, the exercise of self-control in protecting long-term goals from momentary temptations.

Learning Objectives

- Define the basic terminology related to goals, motivation, self-regulation, and self-control.
- Describe the antecedents and consequences of goal activation.
- Describe the factors that influence motivation in the course of goal pursuit.
- Explain the process underlying goal activation, self-regulation, and self-control.
- Give examples of goal activation effects, self-regulation processes, and self-control processes.

Introduction

Every New Year, many people make resolutions—or goals—that go unsatisfied: save money; wake up earlier; go to the gym more. As much as we know our lives would improve if we actually achieved these goals, people quite often don't follow through. But what if that didn't have to be the case? What if every time we made a goal, we actually accomplished it? Each day, our countless small goals guide our decisions and behavior. Even with “mundane” goals,

such as getting the laundry done or showing up to work on time, we are often enacting the same psychological processes involved with achieving loftier dreams. To understand how we can better attain our goals, let's begin with defining what a goal is and what underlies it, psychologically.

A **goal** is the cognitive representation of a desired state, or, in other words, our mental idea of how we'd like things to turn out (Fishbach & Ferguson 2007; Kruglanski, 1996). This desired end state of a goal can be clearly defined (e.g., stepping on the surface of Mars), or it can be more abstract and represent a state that is never fully completed (e.g., eating healthy). Underlying all of these goals, though, is **motivation**, or the psychological driving force that enables action in the pursuit of that goal (Lewin, 1935). Motivation can stem from two places. First, it can come from the benefits associated with the process of pursuing a goal (**intrinsic** motivation). For example, you might be driven by the desire to have a fulfilling experience while working on your Mars mission. Second, motivation can also come from the

benefits associated with achieving a goal (**extrinsic** motivation), such as the fame and fortune that come with being the first person on Mars (Deci & Ryan, 1985). One easy way to consider intrinsic and extrinsic motivation is through the eyes of a student. Does the student work hard on assignments because the act of learning is pleasing (intrinsic motivation)? Or does the student work hard to get good grades, which will help land a good job (extrinsic motivation)?

Social psychologists recognize that goal pursuit and the motivations that underlie it do not depend solely on an individual's personality. Rather, they are products of personal characteristics and situational factors. Indeed, cues in a person's immediate environment—including images, words, sounds, and the presence of other people—can activate, or prime, a goal. This activation can be **conscious**, such that the person is aware of the environmental cues influencing his/her pursuit of a goal. However, this activation can also occur outside a person's awareness, and lead to **nonconscious** goal pursuit. In this case, the person is unaware of why s/he is pursuing a goal and may not even realize that s/he is pursuing it.



Goals are fundamental guides for human behavior. Some are biological in origin, some are cultural in nature and some are unique to the individual. [Image: Thoroughly Reviewed, <https://goo.gl/Dczz2l>, CC BY 2.0, <https://goo.gl/BRvSA7>]

In this module, we review key aspects of goals and motivation. First, we discuss the origins and manifestation of goals. Then, we review factors that influence individuals' motivation in the course of pursuing a goal (**self-regulation**). Finally, we discuss what motivates individuals to keep following their goals when faced with other conflicting desires—for example, when a tempting opportunity to socialize on Facebook presents itself in the course of studying for an exam (**self-control**).

The Origins and Manifestation of Goals

Goal Adoption

What makes us commit to a goal? Researchers tend to agree that commitment stems from the sense that a goal is both valuable and attainable, and that we adopt goals that are highly likely to bring positive outcomes (i.e., one's *commitment* = the *value* of the goal × the *expectancy* it will be achieved) (Fishbein & Ajzen, 1974; Liberman & Förster, 2008). This process of committing to a goal can occur without much conscious deliberation. For example, people infer value and attainability, and will nonconsciously determine their commitment based on those factors, as well as the outcomes of past goals. Indeed, people often learn about themselves the same way they learn about other people—by observing their behaviors (in this case, their own) and drawing inferences about their preferences. For example, after taking a kickboxing class, you might infer from your efforts that you are indeed committed to staying physically fit (Fishbach, Zhang, & Koo, 2009).

Consequences of Goal Activation

The activation of a goal and the accompanying increase in motivation can influence many aspects of behavior and judgment, including how people perceive, evaluate, and feel about the world around them. Indeed, motivational states can



What does this image represent to you, a number or a letter? Training to run the Boston Marathon? Need to pass 13 credit hours to graduate this semester? The details of your goals may influence how you interpret the world around you. [Image: Leo Reynolds, <https://goo.gl/TXhu3C>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

even alter something as fundamental as visual perception. For example, Balcetis and Dunning (2006) showed participants an ambiguous figure (e.g., “13”) and asked them whether they saw the letter B or the number 13. The researchers found that when participants had the goal of seeing a letter (e.g., because seeing a number required the participants to drink a gross tasting juice), they in fact *saw* a B. It wasn’t that the participants were simply lying, either; their goal literally changed how they perceived the world!

Goals can also exert a strong influence on how people evaluate the objects (and people) around them. When pursuing a goal such as quenching one’s thirst, people evaluate goal-relevant objects (e.g., a glass) more positively than objects that are not relevant to the goal (e.g., a pencil). Furthermore, those with the goal of quenching their thirst rate the glass more positively than people who are not pursuing the goal (Ferguson & Bargh, 2004).

Self-Regulation in Goal Pursuit

Many of the behaviors we like to engage in are inconsistent with achieving our goals. For example, a person training for a marathon might also want to stay up late and drink wine on her sister’s wedding day. Self-regulation refers to the process through which individuals alter their perceptions, feelings, and actions in the pursuit of a goal. For example, saying that it is possible to have fun at a wedding even without alcohol might guide the person’s decisions. In the following section, we review the main theories and findings on self-regulation.

From Deliberation to Implementation

Self-regulation involves two basic stages, each with its own distinct mindset. First, a person must decide which of many potential goals to pursue at a given point in time (deliberative phase). While in the deliberative phase, a person often has a mindset that fosters an effective assessment of goals. That is, one tends to be open-minded and realistic about available goals to pursue. However, such scrutiny of one’s choices sometimes hinders action. For example, in the deliberative phase about how to spend time, someone might consider improving health, academic performance, or developing a hobby. At the same time, though, this deliberation involves considering realistic obstacles, such as one’s busy schedule, which may discourage the person from believing the goals can likely be achieved (and thus, doesn’t work toward any of them).

However, after deciding which goal to follow, the second stage involves planning specific actions related to the goal (implemental phase). In the implemental phase, a person tends to have a mindset conducive to the effective implementation of a goal through immediate

action—i.e., with the planning done, we're ready to jump right into attaining our goal. Unfortunately, though, this mindset often leads to closed-mindedness and unrealistically positive expectations about the chosen goal (Gollwitzer, Heckhausen, & Steller, 1990; Kruglanski et al., 2000; Thaler & Shefrin, 1981). For example, in order to follow a health goal, a person might register for a gym membership and start exercising. In doing so, s/he assumes this is all that's needed to achieve the goal (closed-mindedness), and after a few weeks, it should be accomplished (unrealistic expectations).

Regulation of Ought- and Ideals-Goals



Different people might have different orientations toward the same goal. One person - with a prevention orientation - might train for a marathon because they believe it will help with a problematic family health history, while another person - with a promotion orientation - might pursue the same goal in order to feel confident and accomplished. [Image: TheOtherKev, <https://tinyurl.com/m3cvdnya>, <https://tinyurl.com/bdnnhnnkk>]

In addition to two phases in goal pursuit, research also distinguishes between two distinct self-regulatory orientations (or perceptions of effectiveness) in pursuing a goal: **prevention** and **promotion**. A prevention focus emphasizes safety, responsibility, and security needs, and views goals as “oughts.” That is, for those who are *prevention*-oriented, a goal is viewed as something they *should* be doing, and they tend to focus on avoiding potential problems (e.g., exercising to avoid health threats). This self-regulatory focus leads to a vigilant strategy aimed at avoiding losses (the presence of negatives) and approaching non-losses (the absence of negatives). On the other hand, a *promotion* focus views

goals as “ideals,” and emphasizes hopes, accomplishments, and advancement needs. Here, people view their goals as something they *want* to do that will bring them added pleasure (e.g., rock climbing because it is enjoyable to be in nature and to feel a sense of accomplishment). This type of orientation leads to the adoption of an eager strategy concerned with approaching gains (the presence of positives) and avoiding non-gains (the absence of positives).

To compare these two strategies, consider the goal of saving money. Prevention-focused people will save money because they believe it's what they should be doing (an ought), and because they're concerned about not having any money (avoiding a harm). Promotion-focused people, on the other hand, will save money because they want to have extra funds (a desire) so they can do new and fun activities (attaining an advancement). Although these two

strategies result in very similar behaviors, emphasizing potential losses will motivate individuals with a prevention focus, whereas emphasizing potential gains will motivate individuals with a promotion focus. And these orientations—responding better to either a prevention or promotion focus— differ across individuals (chronic regulatory focus) and situations (momentary regulatory focus; Higgins, 1997).

A Cybernetic Process of Self-Regulation

Self-regulation depends on feelings that arise from comparing actual **progress** to expected progress. During goal pursuit, individuals calculate the discrepancy between their current state (i.e., all goal-related actions completed so far) and their desired end state (i.e., what they view as “achieving the goal”). After determining this difference, the person then acts to close that gap (Miller, Galanter, & Pribram, 1960; Powers, 1973). In this cybernetic process of self-regulation (or, internal system directing how a person should control behavior), a higher-than-expected rate of closing the discrepancy creates a signal in the form of positive feelings. For example, if you’re nearly finished with a class project (i.e., a low discrepancy between your progress and what it will take to completely finish), you feel good about yourself. However, these positive feelings tend to make individuals “coast,” or reduce their efforts on the focal goal, and shift their focus to other goals (e.g., you’re almost done with your project for one class, so you start working on a paper for another). By contrast, a lower-than-expected rate of closing the gap elicits negative feelings, which leads to greater effort investment on the focal goal (Carver & Scheier, 1998). If it is the day before a project’s due and you’ve hardly started it, you will likely feel anxious and stop all other activities to make progress on your project.

Highlighting One Goal or Balancing Between Goals

When we’ve completed steps toward achieving our goal, looking back on the behaviors or actions that helped us make such progress can have implications for future behaviors and actions (see The Dynamics of Self-Regulation framework; Fishbach et al., 2009). Remember, **commitment** results from the perceived value and attainability of a goal, whereas *progress* describes the perception of a reduced discrepancy between the current state and desired end state (i.e., the cybernetic process). After achieving a goal, when people interpret their previous actions as a sign of *commitment* to it, they tend to **highlight** the pursuit of that goal, prioritizing it and putting more effort toward it. However, when people interpret their previous actions as a sign of *progress*, they tend to **balance** between the goal and other goals, putting less effort into the focal goal. For example, if buying a product on sale reinforces your *commitment* to the goal of saving money, you will continue to behave financially responsibly. However, if

you perceive the same action (buying the sale item) as evidence of *progress* toward the goal of saving money, you might feel like you can “take a break” from your goal, justifying splurging on a subsequent purchase. Several factors can influence the meanings people assign to previous goal actions. For example, the more confident a person is about a commitment to a goal, the more likely s/he is to infer progress rather than commitment from his/her actions (Koo & Fishbach, 2008).

Conflicting Goals and Self-Control



Immediate gratification has a way of interfering with the pursuit of more significant long-term goals. New shoes feel awfully good right now but don't do anything to get us closer to our financial savings target. [Image: Melanie Allan, <https://goo.gl/gkd8Wf>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

In the pursuit of our ordinary and extraordinary goals (e.g., starting a business, moving overseas, adopting a child), we inevitably come across other goals (e.g., learning accounting, learning a foreign language, learning about child development) that might get in the way of our lofty ambitions. In such situations, we must exercise self-control to stay on course. Self-control is the capacity to control impulses, emotions, desires, and actions in order to resist a temptation (e.g., going on a shopping spree) and protect a valued goal (e.g., stay financially sound). As such, self-control is a process of self-regulation in contexts involving a clear trade-off between long-term interests and some form of immediate gratification (Fishbach & Converse, 2010; Rachlin, 2000; Read, Loewenstein, & Rabin, 1999; Thaler & Shefrin, 1981). For example, whereas

reading each page of a textbook requires self-regulation, doing so while resisting the tempting sounds of friends socializing in the next room requires self-control. And although you may tend to believe self-control is just a personal characteristic that varies across individuals, it is like a muscle, in that it becomes drained by being used but is also strengthened in the process.

Self-Control as an Innate Ability

Mischel, Shoda, and Rodriguez (1989) identified enduring individual differences in self-control

and found that the persistent capacity to postpone immediate gratification for the sake of future interests leads to greater cognitive and social competence over the course of a lifetime. In a famous series of lab experiments (first conducted by Mischel & Baker, 1975), preschoolers 3–5 years old were asked to choose between getting a smaller treat immediately (e.g., a single marshmallow) or waiting as long as 15 minutes to get a better one (e.g., two marshmallows). Some children were better-able to exercise self-control than others, resisting the temptation to take the available treat and waiting for the better one. Following up with these preschoolers ten years later, the researchers found that the children who were able to wait longer in the experiment for the second marshmallow (vs. those who more quickly ate the single marshmallow) performed better academically and socially, and had better psychological coping skills as adolescents.

Self-Control as a Limited Resource

Beyond personal characteristics, the ability to exercise self-control can fluctuate from one context to the next. In particular, previous exertion of self-control (e.g., resisting the temptation to splurge on a new computer) drains individuals of the limited physiological and psychological resources required to continue the pursuit of a goal (e.g., later in the day, resisting another tempting purchase). Ego-depletion refers to this exhaustion of resources from resisting a temptation. That is, just like bicycling for two hours would exhaust someone before a basketball game, exerting self-control reduces individuals' capacity to exert more self-control in a consequent task—whether that task is in the same domain (e.g., resisting a donut and then continuing to eat healthy) or a



Willpower is limited. Trying to resist temptation now takes energy and may leave you feeling like it's harder to be disciplined later. You can only eat so many radishes . . . [Image: Dale Cruse, <https://goo.gl/cl0Nox>, CC BY 2.0, <https://goo.gl/BRvSA7>]

different one (e.g., resisting a donut and then continuing to be financially responsible; Baumeister, Bratslavsky, Muraven, & Tice, 1998; Vohs & Heatherton, 2000). For example, in a study by Baumeister et al. (1998), research participants who forced themselves to eat radishes instead of tempting chocolates were subsequently less persistent (i.e., gave up sooner) at attempting an unsolvable puzzle task compared to the participants who had not exerted self-

control to resist the chocolates.

A Prerequisite to Self-Control: Identification

Although factors such as resources and personal characteristics contribute to the successful exercise of self-control, identifying the self-control conflict inherent to a particular situation is an important—and often overlooked—prerequisite. For example, if you have a long-term goal of getting better sleep but don't perceive that staying up late on a Friday night is inconsistent with this goal, you won't have a self-control conflict. The successful pursuit of a goal in the face of temptation requires that individuals first identify they are having impulses that need to be controlled. However, individuals often fail to identify self-control conflicts because many everyday temptations seem to have very minimal negative consequences: one frivolous purchase is unlikely to destroy a person's college savings, but what about 200 frivolous purchases over the course of a few months?

People are more likely to identify a self-control conflict, and exercise self-control, when they think of a choice as part of a broader pattern of repeated behavior rather than as an isolated choice. For example, rather than viewing one's lateness to class as an isolated behavioral decision, the person should try to recognize that this may actually be part of an on-going habit of lying in bed and scrolling on one's phone before class. Indeed, when considering broader decision patterns, consistent temptations become more problematic for long-term interests (Rachlin, 2000; Read, Loewenstein, & Kalyanaraman, 1999). Moreover, conflict identification is more likely if people see their current choices as similar to their future choices.

Self-Control Processes: Counteracting Temptation

The protection of a valued goal involves several cognitive and behavioral strategies ultimately aimed at "counteracting" the pull of temptations and pushing oneself toward goal-related alternatives (Fishbach & Trope, 2007). One such cognitive process involves decreasing the value of temptations and increasing the value of goal-consistent objects or actions. For example, person who prizes academic performance might tell themselves that seeing checking Tik Tok is less appealing than learning about the way that the economy works. Other behavioral strategies include a precommitment to pursue goals and forgo temptation (e.g., leaving one's credit card at home before going to the mall), establishing rewards for goals and penalties for temptations, or physically approaching goals and distancing oneself from temptations (e.g., not downloading time-wasting games onto one's phone). These self-control processes can benefit individuals' long-term interests, either consciously or without conscious awareness. Thus, at times, individuals automatically activate goal-related thoughts in response

to temptation, and inhibit temptation-related thoughts in the presence of goal cues (Fishbach, Friedman, & Kruglanski, 2003).

Conclusion

People often make New Year's resolutions with the idea that attaining one's goals is simple: "I just have to choose to eat healthier, right?" However, after going through this module and learning a social-cognitive approach to the main theories and findings on goals and motivation, we see that even the most basic decisions take place within a much larger and more complex mental framework. Looking back on prior goal failures, it may seem impossible to achieve some of our desires. But, through understanding our own mental representation of our goals (i.e., the values and expectancies behind them), we can help cognitively modify our behavior to achieve our dreams.

Discussion Questions

1. What is the difference between goal and motivation?
2. What is the difference between self-regulation and self-control?
3. How do positive and negative feelings inform goal pursuit in a cybernetic self-regulation process?
4. Describe the characteristics of the deliberative mindset that allows individuals to decide between different goals. How might these characteristics hinder the implemental phase of self-regulation?
5. You just read a module on “Goals and Motivation,” and you believe it is a sign of commitment to the goal of learning about social psychology. Define commitment in this context. How would interpreting your efforts as a sign of commitment influence your motivation to read more about social psychology? By contrast, how would interpreting your efforts as a sign of progress influence your motivation to read more?
6. Mel and Alex are friends. Mel has a prevention focus self-regulatory orientation, whereas Alex has a promotion focus. They are both training for a marathon and are looking for motivational posters to hang in their respective apartments. While shopping, they find a poster with the following Confucius quote: “The will to win, the desire to succeed, the urge to reach your full potential These are the keys that will unlock the door to personal excellence.” Who is this poster more likely to help stay motivated for the marathon (Mel or Alex)? Why? Find or write a quote that might help the other friend.
7. Give an example in which an individual fails to exercise self-control. What are some factors that can cause such a self-control failure?

Vocabulary

Balancing between goals

Shifting between a focal goal and other goals or temptations by putting less effort into the focal goal—usually with the intention of coming back to the focal goal at a later point in time.

Commitment

The sense that a goal is both valuable and attainable

Conscious goal activation

When a person is fully aware of contextual influences and resulting goal-directed behavior.

Deliberative phase

The first of the two basic stages of self-regulation in which individuals decide which of many potential goals to pursue at a given point in time.

Ego-depletion

The exhaustion of physiological and/or psychological resources following the completion of effortful self-control tasks, which subsequently leads to reduction in the capacity to exert more self-control.

Extrinsic motivation

Motivation stemming from the benefits associated with achieving a goal such as obtaining a monetary reward.

Goal

The cognitive representation of a desired state (outcome).

Highlighting a goal

Prioritizing a focal goal over other goals or temptations by putting more effort into the focal goal.

Implemental phase

The second of the two basic stages of self-regulation in which individuals plan specific actions related to their selected goal.

Intrinsic motivation

Motivation stemming from the benefits associated with the process of pursuing a goal such

as having a fulfilling experience.

Motivation

The psychological driving force that enables action in the course of goal pursuit.

Nonconscious goal activation

When activation occurs outside a person's awareness, such that the person is unaware of the reasons behind her goal-directed thoughts and behaviors.

Prevention focus

One of two self-regulatory orientations emphasizing safety, responsibility, and security needs, and viewing goals as "oughts." This self-regulatory focus seeks to avoid losses (the presence of negatives) and approach non-losses (the absence of negatives).

Progress

The perception of reducing the discrepancy between one's current state and one's desired state in goal pursuit.

Promotion focus

One of two self-regulatory orientations emphasizing hopes, accomplishments, and advancement needs, and viewing goals as "ideals." This self-regulatory focus seeks to approach gains (the presence of positives) and avoid non-gains (the absence of positives).

Self-control

The capacity to control impulses, emotions, desires, and actions in order to resist a temptation and adhere to a valued goal.

Self-regulation

The processes through which individuals alter their emotions, desires, and actions in the course of pursuing a goal.

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Drive States

Sudeep Bhatia & George Loewenstein

Our thoughts and behaviors are strongly influenced by affective experiences known as drive states. These drive states motivate us to fulfill goals that are beneficial to our survival and reproduction. This module provides an overview of key drive states, including information about their neurobiology and their psychological effects.

Learning Objectives

- Identify the key properties of drive states
- Describe biological goals accomplished by drive states
- Give examples of drive states
- Outline the neurobiological basis of drive states such as hunger and arousal
- Discuss the main moderators and determinants of drive states such as hunger and arousal

Introduction

What is the longest you've ever gone without eating? A couple of hours? An entire day? How did it feel? Humans rely critically on food for nutrition and energy, and the absence of food can create drastic changes, not only in physical appearance, but in thoughts and behaviors. If you've ever fasted for a day, you probably noticed how hunger can take over your mind, directing your attention to foods you could be eating (a cheesy slice of pizza, or perhaps some sweet, cold ice cream), and motivating you to obtain and consume these foods. And once you have eaten and your hunger has been satisfied, your thoughts and behaviors return to normal.



Hunger is among our most basic motivators. [Image: Jeremy Brooks, <https://goo.gl/XrFG2W>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

Hunger is a **drive state**, an affective experience (something you feel, like the sensation of being tired or hungry) that motivates organisms to fulfill goals that are generally beneficial to their survival and reproduction. Like other drive states, such as thirst or sexual arousal, hunger has a profound impact on the functioning of the mind. It affects psychological processes, such as perception, attention, emotion, and motivation, and influences the behaviors that these processes generate.

Key Properties of Drive States

Drive states differ from other affective or emotional states in terms of the biological

functions they accomplish. Whereas all affective states possess valence (i.e., they are positive or negative) and serve to motivate approach or avoidance behaviors (Zajonc, 1998), drive states are unique in that they generate behaviors that result in specific benefits for the body. For example, hunger directs individuals to eat foods that increase blood sugar levels in the body, while thirst causes individuals to drink fluids that increase water levels in the body.

Different drive states have different triggers. Most drive states respond to both internal and external cues, but the combinations of internal and external cues, and the specific types of cues, differ between drives. Hunger, for example, depends on internal, visceral signals as well as sensory signals, such as the sight or smell of tasty food. Different drive states also result in different cognitive and emotional states, and are associated with different behaviors. Yet despite these differences, there are a number of properties common to all drive states.

Homeostasis

Humans, like all organisms, need to maintain a stable state in their various physiological systems. For example, the excessive loss of body water results in dehydration, a dangerous and potentially fatal state. However, too much water can be damaging as well. Thus, a moderate and stable level of body fluid is ideal. The tendency of an organism to maintain this stability across all the different physiological systems in the body is called **homeostasis**.

Homeostasis is maintained via two key factors. First, the state of the system being regulated must be monitored and compared to an ideal level, or a **set point**. Second, there need to be mechanisms for moving the system back to this set point—that is, to restore homeostasis when deviations from it are detected. To better understand this, think of the thermostat in your own home. It detects when the current temperature in the house is different than the temperature you have it set at (i.e., the set point). Once the thermostat recognizes the difference, the heating or air conditioning turns on to bring the overall temperature back to the designated level.

Many homeostatic mechanisms, such as blood circulation and immune responses, are automatic and nonconscious. Others, however, involve deliberate action. Most drive states motivate action to restore homeostasis using both “punishments” and “rewards.” Imagine that these homeostatic mechanisms are like molecular parents. When you behave poorly by departing from the set point (such as not eating or being somewhere too cold), they raise their voice at you. You experience this as the bad feelings, or “punishments,” of hunger, thirst, or feeling too cold or too hot. However, when you behave well (such as eating nutritious foods when hungry), these homeostatic parents reward you with the pleasure that comes from any activity that moves the system back toward the set point. For example, when body temperature declines below the set point, any activity that helps to restore homeostasis (such as putting one’s hand in warm water) feels pleasurable; and likewise, when body temperature rises above the set point, anything that cools it feels pleasurable.



The body needs homeostasis and motivates us - through both pleasure and pain - to stay in balance. [Image: ashleigh290, <https://goo.gl/yXQtEC>, CC-BY 2.0, <https://goo.gl/BRvSA7>]

The Narrowing of Attention

As drive states intensify, they direct attention toward elements, activities, and forms of consumption that satisfy the biological needs associated with the drive. Hunger, for example, draws attention toward food. Outcomes and objects that are not related to satisfying hunger

lose their value (Easterbrook, 1959). For instance, has anyone ever invited you to do a fun activity while you were hungry? Likely your response was something like: “I’m not doing anything until I eat first.” Indeed, at a sufficient level of intensity, individuals will sacrifice almost any quantity of goods that do not address the needs signaled by the drive state. For example, cocaine addicts, according to Gawin (1991:1581), “report that virtually all thoughts are focused on cocaine during binges; nourishment, sleep, money, loved ones, responsibility, and survival lose all significance.”

Drive states also produce a second form of attention-narrowing: a collapsing of time-perspective toward the present. That is, they make us impatient. While this form of attention-narrowing is particularly pronounced for the outcomes and behaviors directly related to the biological function being served by the drive state at issue (e.g., “I need food *now*”), it applies to general concerns for the future as well. Ariely and Loewenstein (2006), for example, investigated the impact of sexual arousal on the thoughts and behaviors of a sample of male undergraduates. These undergraduates were lent laptop computers that they took to their private residences, where they answered a series of questions, both in normal states and in states of high sexual arousal. Ariely and Loewenstein found that being sexually aroused made people extremely impatient for both sexual outcomes and for outcomes in other domains, such as those involving money. In another study Giordano et al. (2002) found that heroin addicts were more impatient with respect to heroin when they were craving it than when they were not. More surprisingly, they were also more impatient toward money (they valued delayed money less) when they were actively craving heroin.

Yet a third form of attention-narrowing involves thoughts and outcomes related to the self versus others. Intense drive states tend to narrow one’s focus inwardly and to undermine altruism—or the desire to do good for others. People who are hungry, in pain, or craving drugs tend to be selfish. Indeed, popular interrogation methods involve depriving individuals of sleep, food, or water, so as to trigger intense drive states leading the subject of the interrogation to divulge information that may betray comrades, friends, and family (Biderman, 1960).

Two Illustrative Drive States

Thus far we have considered drive states abstractly. We have discussed the ways in which they relate to other affective and motivational mechanisms, as well as their main biological purpose and general effects on thought and behavior. Yet, despite serving the same broader goals, different drive states are often remarkably different in terms of their specific properties. To understand some of these specific properties, we will explore two different drive states

Current Controversy

In 2005, the American Psychological Association (APA) issued a report concluding that psychologists could ethically play a role in the interrogation of people captured in Afghanistan and elsewhere. In 2014, following critical media publicity documenting the APA's involvement in torture, the APA commissioned a law firm to independently investigate APA involvement in interrogation. The firm's report was damaging to the APA because it suggested that APA leaders colluded with the Department of Defense, CIA, and other government officials not only to aid in interrogation itself, but to provide justification for government guidelines that defined torture (which is banned by international treaties signed by the U.S.) in a narrow fashion that excluded, for example so-called "stress positions" and sleep deprivation.

Critical Questions

1. Do you think that manipulating drive states, such as the need for sleep, constitutes torture?
2. How do you think research on drive states should inform the definition of "torture" and our definition of ethical interrogation techniques?

See the full Hoffman Report here - <http://www.apa.org/independent-review/APA-FINAL-Report-7.2.15.pdf>

For more coverage - <http://www.nytimes.com/2015/07/11/us/psychologists-shielded-us-torture-program-report-finds.html>

that play very important roles in determining behavior, and in ensuring human survival: hunger and sexual arousal.

Hunger

Hunger is a classic example of a drive state, one that results in thoughts and behaviors related to the consumption of food. Hunger is generally triggered by low glucose levels in the blood (Rolls, 2000), and behaviors resulting from hunger aim to restore homeostasis regarding those glucose levels. Various other internal and external cues can also cause hunger. For example, when fats are broken down in the body for energy, this initiates a chemical cue that the body should search for food (Greenberg, Smith, & Gibbs, 1990). External cues include the time of day, estimated time until the next feeding (hunger increases immediately prior to food consumption), and the sight, smell, taste, and even touch of food and food-related stimuli. Note that while hunger is a generic feeling, it has nuances that can provoke the eating of specific foods that correct for nutritional imbalances we may not even be conscious of. For



External cues, like the sight and smell of food, can ignite feelings of hunger. [Image: Marco Verch, <https://goo.gl/c4TC5A>, CC BY 2.0, <https://goo.gl/BRvSA7>]

example, a couple who was lost adrift at sea found they inexplicably began to crave the eyes of fish. Only later, after they had been rescued, did they learn that fish eyes are rich in vitamin C—a very important nutrient that they had been depleted of while lost in the ocean (Walker, 2014).

The hypothalamus (located in the lower, central part of the brain) plays a very important role in eating behavior. It is responsible for synthesizing and secreting various hormones. The lateral hypothalamus (LH) is concerned largely with hunger and, in fact, lesions (i.e., damage) of the LH can eliminate the desire for eating entirely—to the point that animals starve themselves to death unless kept alive by force feeding

(Anand & Brobeck, 1951). Additionally, artificially stimulating the LH, using electrical currents, can generate eating behavior if food is available (Andersson, 1951).

Activation of the LH can not only increase the desirability of food but can also reduce the desirability of nonfood-related items. For example, Brendl, Markman, and Messner (2003) found that participants who were given a handful of popcorn to trigger hunger not only had higher ratings of food products, but also had lower ratings of nonfood products—compared with participants whose appetites were not similarly primed. That is, because eating had become more important, other non-food products lost some of their value.

Hunger is only part of the story of when and why we eat. A related process, satiation, refers to the decline of hunger and the eventual termination of eating behavior. Whereas the feeling of hunger gets you to start eating, the feeling of satiation gets you to stop. Perhaps surprisingly, hunger and satiation are two distinct processes, controlled by different circuits in the brain and triggered by different cues. Distinct from the LH, which plays an important role in hunger, the ventromedial hypothalamus (VMH) plays an important role in satiety. Though lesions of the VMH can cause an animal to overeat to the point of obesity, the relationship between the LH and the VMB is quite complicated. Rats with VMH lesions can also be quite finicky about their food (Teitelbaum, 1955).

Other brain areas, besides the LH and VMH, also play important roles in eating behavior. The

sensory cortices (visual, olfactory, and taste), for example, are important in identifying food items. These areas provide informational value, however, not hedonic evaluations. That is, these areas help tell a person what is good or safe to eat, but they don't provide the pleasure (or hedonic) sensations that *actually* eating the food produces. While many sensory functions are roughly stable across different psychological states, other functions, such as the detection of food-related stimuli, are enhanced when the organism is in a hungry drive state.

After identifying a food item, the brain also needs to determine its **reward value**, which affects the organism's motivation to consume the food. The reward value ascribed to a particular item is, not surprisingly, sensitive to the level of hunger experienced by the organism. The hungrier you are, the greater the reward value of the food. Neurons in the areas where reward values are processed, such as the orbitofrontal cortex, fire more rapidly at the sight or taste of food when the organism is hungry relative to if it is satiated.

Sexual Arousal

A second drive state, especially critical to reproduction, is sexual arousal. Sexual arousal results in thoughts and behaviors related to sexual activity. As with hunger, it is generated by a large range of internal and external mechanisms that are triggered either after the extended absence of sexual activity or by the immediate presence and possibility of sexual activity (or by cues commonly associated with such possibilities). Unlike hunger, however, these mechanisms can differ substantially between males and females, indicating important evolutionary differences in the biological functions that sexual arousal serves for different sexes.

Sexual arousal and pleasure in males, for example, is strongly related to the **preoptic area**, a region in the anterior hypothalamus (or the front of the hypothalamus). If the preoptic area is damaged, male sexual behavior is severely impaired. For example, rats that have had prior sexual experiences will still seek out sexual partners after their preoptic area is lesioned.



Unlike other drive states the mechanisms that trigger sexual arousal are not the same for men and women. [Image: Matthew Romack, <https://goo.gl/IUbbk0>, CC BY 2.0, <https://goo.gl/BRvSA7>]

However, once having secured a sexual partner, rats with lesioned preoptic areas will show no further inclination to actually initiate sex.

For females, though, the preoptic area fulfills different roles, such as functions involved with eating behaviors. Instead, there is a different region of the brain, the ventromedial hypothalamus (the lower, central part) that plays a similar role for females as the preoptic area does for males. Neurons in the ventromedial hypothalamus determine the excretion of estradiol, an estrogen hormone that regulates sexual receptivity (or the willingness to accept a sexual partner). In many mammals, these neurons send impulses to the periaqueductal gray (a region in the midbrain) which is responsible for defensive behaviors, such as freezing immobility, running, increases in blood pressure, and other motor responses. Typically, these defensive responses might keep the female rat from interacting with the male one. However, during sexual arousal, these defensive responses are weakened and lordosis behavior, a physical sexual posture that serves as an invitation to mate, is initiated (Kow and Pfaff, 1998). Thus, while the preoptic area encourages males to engage in sexual activity, the ventromedial hypothalamus fulfills that role for females.

Other differences between males and females involve overlapping functions of neural modules. These neural modules often provide clues about the biological roles played by sexual arousal and sexual activity in males and females. Areas of the brain that are important for male sexuality overlap to a great extent with areas that are also associated with aggression. In contrast, areas important for female sexuality overlap extensively with those that are also connected to nurturance (Panksepp, 2004).

One region of the brain that seems to play an important role in sexual pleasure for both males and females is the septal nucleus, an area that receives reciprocal connections from many other brain regions, including the hypothalamus and the amygdala (a region of the brain primarily involved with emotions). This region shows considerable activity, in terms of rhythmic spiking, during sexual orgasm. It is also one of the brain regions that rats will most reliably voluntarily self-stimulate (Olds & Milner, 1954). In humans, placing a small amount of acetylcholine into this region, or stimulating it electrically, has been reported to produce a feeling of imminent orgasm (Heath, 1964).

Conclusion

Drive states are evolved motivational mechanisms designed to ensure that organisms take self-beneficial actions. In this module, we have reviewed key properties of drive states, such as homeostasis and the narrowing of attention. We have also discussed, in some detail, two

important drive states—hunger and sexual arousal—and explored their underlying neurobiology and the ways in which various environmental and biological factors affect their properties.

There are many drive states besides hunger and sexual arousal that affect humans on a daily basis. Fear, thirst, exhaustion, exploratory and maternal drives, and drug cravings are all drive states that have been studied by researchers (see e.g., Buck, 1999; Van Boven & Loewenstein, 2003). Although these drive states share some of the properties discussed in this module, each also has unique features that allow it to effectively fulfill its evolutionary function.

One key difference between drive states is the extent to which they are triggered by internal as opposed to external stimuli. Thirst, for example, is induced both by decreased fluid levels and an increased concentration of salt in the body. Fear, on the other hand, is induced by perceived threats in the external environment. Drug cravings are triggered both by internal homeostatic mechanisms and by external visual, olfactory, and contextual cues. Other drive states, such as those pertaining to maternity, are triggered by specific events in the organism's life. Differences such as these make the study of drive states a scientifically interesting and important endeavor. Drive states are rich in their diversity, and many questions involving their neurocognitive underpinnings, environmental determinants, and behavioral effects, have yet to be answered.

One final thing to consider, not discussed in this module, relates to the real-world consequences of drive states. Hunger, sexual arousal, and other drive states are all psychological mechanisms that have evolved gradually over millions of years. We share these drive states not only with our human ancestors but with other animals, such as monkeys, dogs, and rats. It is not surprising then that these drive states, at times, lead us to behave in ways that are ill-suited to our modern lives. Consider, for example, the obesity epidemic that is affecting countries around the world. Like other diseases of affluence, obesity is a product of drive states that are too easily fulfilled: homeostatic mechanisms that once worked well when food was scarce now backfire when meals rich in fat and sugar are readily available. Unrestricted sexual arousal can have similarly perverse effects on our well-being. Countless politicians have sacrificed their entire life's work (not to mention their marriages) by indulging adulterous sexual impulses toward colleagues, staffers, sex workers, and others over whom they have social or financial power. It is not an overstatement to say that many problems of the 21st century, from school massacres to obesity to drug addiction, are influenced by the mismatch between our drive states and our uniquely modern ability to fulfill them at a moment's notice.

Outside Resources

Web: An open textbook chapter on homeostasis

http://en.wikibooks.org/wiki/Human_Physiology/Homeostasis

Web: Motivation and emotion in psychology

http://allpsych.com/psychology101/motivation_emotion.html

Web: The science of sexual arousal

<http://www.apa.org/monitor/apr03/arousal.aspx>

Discussion Questions

1. The ability to maintain homeostasis is important for an organism's survival. What are the ways in which homeostasis ensures survival? Do different drive states accomplish homeostatic goals differently?
2. Drive states result in the narrowing of attention toward the present and toward the self. Which drive states lead to the most pronounced narrowing of attention toward the present? Which drive states lead to the most pronounced narrowing of attention toward the self?
3. What are important differences between hunger and sexual arousal, and in what ways do these differences reflect the biological needs that hunger and sexual arousal have been evolved to address?
4. Some of the properties of sexual arousal vary across males and females. What other drives states affect males and females differently? Are there drive states that vary with other differences in humans (e.g., age)?

Vocabulary

Drive state

Affective experiences that motivate organisms to fulfill goals that are generally beneficial to their survival and reproduction.

Homeostasis

The tendency of an organism to maintain a stable state across all the different physiological systems in the body.

Homeostatic set point

An ideal level that the system being regulated must be monitored and compared to.

Hypothalamus

A portion of the brain involved in a variety of functions, including the secretion of various hormones and the regulation of hunger and sexual arousal.

Lordosis

A physical sexual posture in females that serves as an invitation to mate.

Preoptic area

A region in the anterior hypothalamus involved in generating and regulating male sexual behavior.

Reward value

A neuropsychological measure of an outcome's affective importance to an organism.

Satiation

The state of being full to satisfaction and no longer desiring to take on more.

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